ESAT PRACTICE TEST SERIES

PHYSICS + CHEMISTRY

LEVEL 2 · HARDER THAN ESAT

Physics + Chemistry

Mathematics 1 + Physics + Chemistry

Challenge material with denser interpretation, quantitative reasoning and longer solution chains.

MODULE
Mathematics 1

MODULE
Physics

MODULE
Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Physics + Chemistry · Level 2 Harder than ESAT

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Physics + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 5 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 D	Q4 A	Q5 E
Q6 C	Q7 B	Q8 D	Q9 A	Q10 E
Q11 C	Q12 B	Q13 D	Q14 A	Q15 E
Q16 B	Q17 C	Q18 D	Q19 A	Q20 E
Q21 C	Q22 B	Q23 D	Q24 A	Q25 E
Q26 C	Q27 B			

Physics

Engineering Mock Test 5 · preserved Physics module

Q1 D	Q2 D	Q3 C	Q4 C	Q5 B
Q6 D	Q7 D	Q8 C	Q9 D	Q10 C
Q11 E	Q12 D	Q13 C	Q14 C	Q15 D
Q16 D	Q17 E	Q18 D	Q19 C	Q20 C
Q21 B	Q22 B	Q23 D	Q24 D	Q25 B
Q26 C	Q27 C			

Chemistry

Preserved Chemistry Harder than ESAT source set

Q1 A	Q2 D	Q3 F	Q4 H	Q5 C
Q6 G	Q7 B	Q8 G	Q9 E	Q10 F
Q11 C	Q12 B	Q13 E	Q14 A	Q15 C
Q16 D	Q17 C	Q18 B	Q19 E	Q20 B
Q21 A	Q22 D	Q23 F	Q24 D	Q25 F
Q26 E	Q27 A			

LEVEL 2 · HARDER THAN ESAT

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Laws of indices and exact simplification

Answer C

QUESTION 1

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\dfrac{10y^3\sqrt{y}}{x}$

B $\dfrac{50y^3}{x}$

C $\dfrac{50y^3\sqrt{y}}{x}$

D $\dfrac{50\sqrt{y}}{x}$

E $\dfrac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2}$ and $(2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}$.

So the numerator is $2500x^{3}y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}$.

The denominator is $(20x^{1/2}y)^{2} = 400xy^{2}$.

Therefore

$$\frac{20000y^{11/2}}{400xy^{2}} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^{3}\sqrt{y}}{x}}.$$

M1-02

Successive percentage change and comparison

Answer B

QUESTION 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q?

A $28 : 33$

B $33 : 28$

C $11 : 10$

D $6 : 5$

E $99 : 80$

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

M1-03

Algebraic fractions and hidden substitution

Answer D

QUESTION 3

For real x, define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{13}{2},$$

what is the value of x?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify first:

$$E = \frac{\frac{(x+2)-(x-3)}{(x-3)(x+2)}}{\frac{(x+2)+(x-3)}{(x-3)(x+2)}} = \frac{5}{2x-1}.$$

Let $t = E$. Then $t + \frac{1}{t} = \frac{13}{2}$, so

$$2t^2 - 13t + 2 = 0 = (2t-1)(t-6).$$

Hence $t = 6$ or $t = \frac{1}{6}$. Since $x > 3$, we have $0 < E < 1$, so $E = \frac{1}{6}$.

Therefore

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow 2x - 1 = 30 \Rightarrow \boxed{x = \frac{31}{2}}.$$

M1-04 Inequalities with domain restrictions **Answer A**

QUESTION 4

How many integer values of x satisfy all three conditions?

- $-5 \le x \le 5$
- $\frac{x^2 + 3x - 10}{x^2 - 1}$ is defined
- $\frac{x^2 + 3x - 10}{x^2 - 1} \le 1$

A 6

B 5

C 4

D 7

E 8

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \qquad x \neq \pm 1.$$

A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are $\boxed{6}$ values.

M1-05

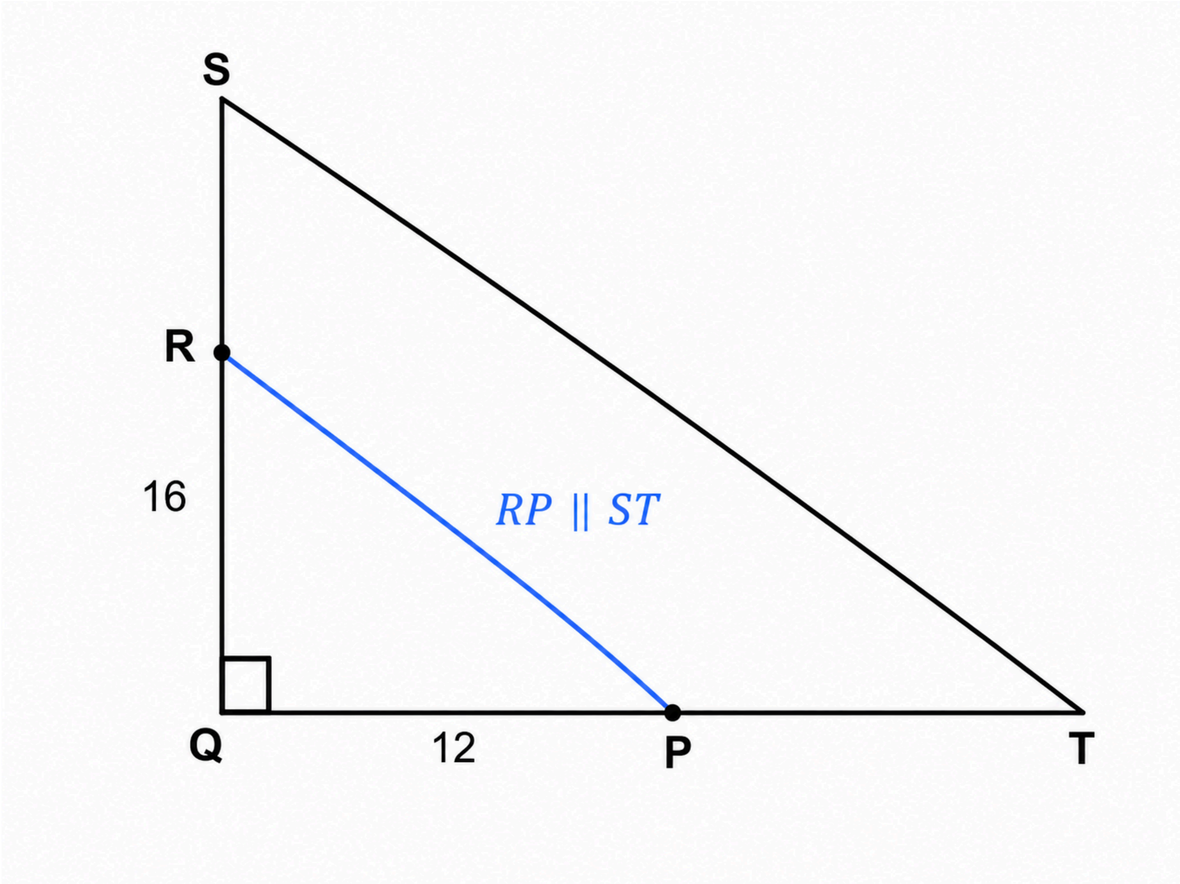
Similarity and area scaling in a right triangle

Answer **E**

QUESTION 5

Triangle SQT is right-angled at Q, with $SQ = 16$ cm and $QT = 12$ cm.

Point R lies on SQ such that $SR : RQ = 1 : 3$. Through R, a line parallel to ST meets QT at P.



What is the area of quadrilateral $SRPT$?

A $24\ \mathrm{cm}^2$

B $30\ \mathrm{cm}^2$

C $36\ \mathrm{cm}^2$

D $40\ \mathrm{cm}^2$

E $42\ \mathrm{cm}^2$

WORKED SOLUTION

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96.$

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12.$

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor $\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}$.

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42 \text{ cm}^2}$.

M1-06

Arithmetic sequences and common terms

Answer **C**

QUESTION 6

The nth term of sequence T is $4n - 3$.

The nth term of sequence V is $7n + p$, where p is an integer and $0 \leq p < 7$.

The sequences have consecutive common terms 53 and 81.

What is the value of p?

- **A** 1
- **B** 2
- **C** 4
- **D** 5
- **E** 6

WORKED SOLUTION

Using the common term 53,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod{7}$, and $0 \le p < 7$, we must have $\boxed{p = 4}$.

M1-07

Cylinders with swapped dimensions

Answer B

QUESTION 7

Two solid cylinders P and Q are defined using positive numbers x and y, where $x > y$.

- Cylinder P has diameter x and height y.
- Cylinder Q has diameter y and height x.

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q.

The volume of P exceeds the volume of Q by $32\pi\ \text{cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h, total surface area is $2\pi(\frac{d}{2})^2 + 2\pi(\frac{d}{2})h$.

Thus

$$A_P = \pi\left(\frac{x^2}{2} + xy\right), \qquad A_Q = \pi\left(\frac{y^2}{2} + xy\right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

M1-08

Ratio change after equal spending

Answer **D**

QUESTION 8

Pat and Alex have a combined total of £ 84. The ratio of Pat's money to Alex's money is $5 : 2$.

They each spend the same amount on sweets.

After this, Pat receives £ 6 and Alex receives £ 1. The ratio of their new amounts is then $3 : 1$.

How much did each of them spend?

A £ 3

B £ 4

C £ 4.25

D £ 4.50

E £ 6

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s, then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent $\boxed{£\ 4.50}$.

M1-09

Coordinate geometry of a tangent

Answer A

QUESTION 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E $y = \frac{4}{3}x + \frac{17}{3}$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \Leftrightarrow \boxed{3x - 4y + 25 = 0}.$$

M1-10 Conditional probability with non-standard dice **Answer E**

QUESTION 10

Two fair dice X and Y are available.

- The faces on X are labelled $1, 1, 2, 3, 4, 5$.
- The faces on Y are labelled $2, 3, 3, 4, 5, 6$.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X?

A $\frac{1}{3}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X, the ordered outcomes summing to 7 are $(2,5),(5,2),(3,4),(4,3)$: 4 outcomes out of 36. So $P(7 \mid X) = \frac{1}{9}$.

For die Y, 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence $P(7 \mid Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X \mid 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

M1-11 Exponential equation via substitution Answer **C**

QUESTION 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t-1)(t-4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is $\boxed{2}$.

M1-12 Rearranging to make a variable the subject **Answer B**

QUESTION 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x?

A $\dfrac{x - \sqrt{x^2 - 16}}{2}$

B $\dfrac{x + \sqrt{x^2 - 16}}{2}$

C $\dfrac{x + \sqrt{x^2 + 16}}{2}$

D $\dfrac{2}{x + \sqrt{x^2 - 16}}$

E $\dfrac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y:

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$\boxed{y = \frac{x + \sqrt{x^2 - 16}}{2}}.$$

M1-13 Surface area comparison of a cylinder and a cube **Answer D**

QUESTION 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi}\, r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\dfrac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder’s total surface area is $2\pi r^2 + 2\pi r h = 2\pi r(r + h)$.

The cube’s total surface area is $6(\sqrt{\pi} r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

M1-14

Mean and median with an ordered list

Answer A

QUESTION 14

Consider the list

$$1,\ 2,\ x,\ x + 1,\ 2x - 1,\ 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

M1-15

Speed, time and average-rate reasoning

Answer E

QUESTION 15

A cyclist rides from A to B uphill at 12 km h^{-1}, immediately turns round, and returns from B to A along the same route at 18 km h^{-1}.

She spends exactly 6 minutes stationary at B.

The total time for the complete trip is 46 minutes.

What is the distance from A to B?

A 3.6 km

B 4.0 km

C 4.2 km

D 4.5 km

E 4.8 km

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{d}{12} + \frac{d}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$,

$$d \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow d = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is $\boxed{4.8\text{ km}}$.

M1-16

Exact trigonometric values

Answer **B**

QUESTION 16

Given that

$$y = \frac{2\sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2?

A $\dfrac{4 - 2\sqrt{3}}{3}$

B $\dfrac{8 - 4\sqrt{3}}{3}$

C $\dfrac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \qquad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6}-\sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6}-\sqrt{2})^2}{3} = \frac{6+2-2\sqrt{12}}{3} = \frac{8-4\sqrt{3}}{3}.$$

M1-17

Inverse proportion in standard form

Answer C

QUESTION 17

The variable t is inversely proportional to the square of w.

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5}.

What is the value of w when $t = 8 \times 10^{-6}$?

A 7.5×10^3

B $3\sqrt{10} \times 10^2$

C $1.5\sqrt{10} \times 10^3$

D $2\sqrt{10} \times 10^3$

E 4.5×10^3

WORKED SOLUTION

Since $t = \frac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

M1-18

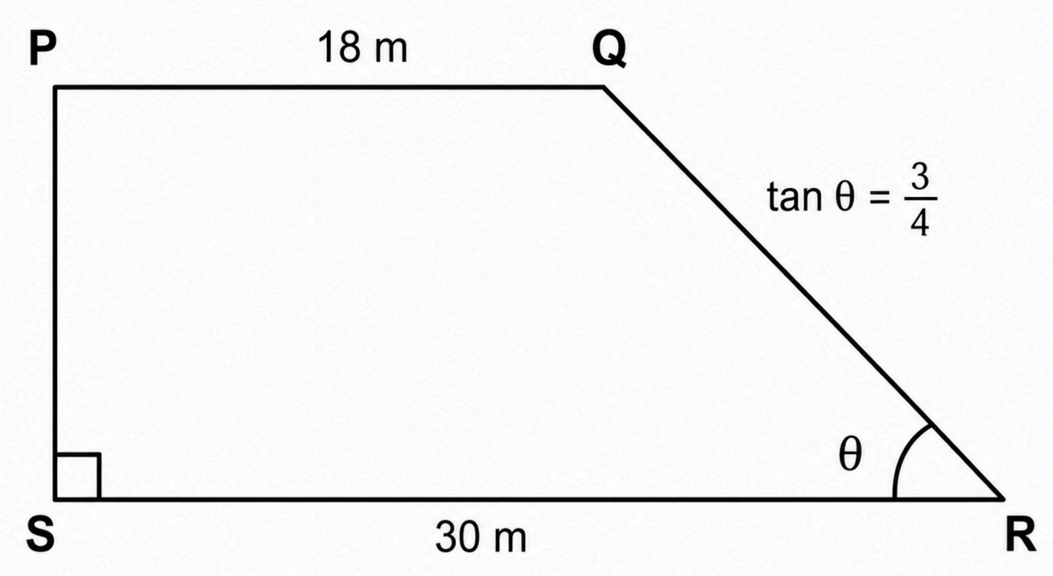
Trapezium area using trigonometry

Answer **D**

QUESTION 18

$PQRS$ is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18$ m, $SR = 30$ m, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A $180\ \text{m}^2$

B $192\ \mathrm{m}^2$

C $204\ \mathrm{m}^2$

D $216\ \mathrm{m}^2$

E $240\ \mathrm{m}^2$

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h, then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216\ \mathrm{m}^2}.$$

M1-19

Quadratic roots and discriminant reasoning

Answer A

QUESTION 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s, then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence $\boxed{b = 6}$.

M1-20 Algebraic geometry with area change Answer **E**

QUESTION 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E 30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is $2(9 + 6) = \boxed{30 \text{ cm}}$.

M1-21

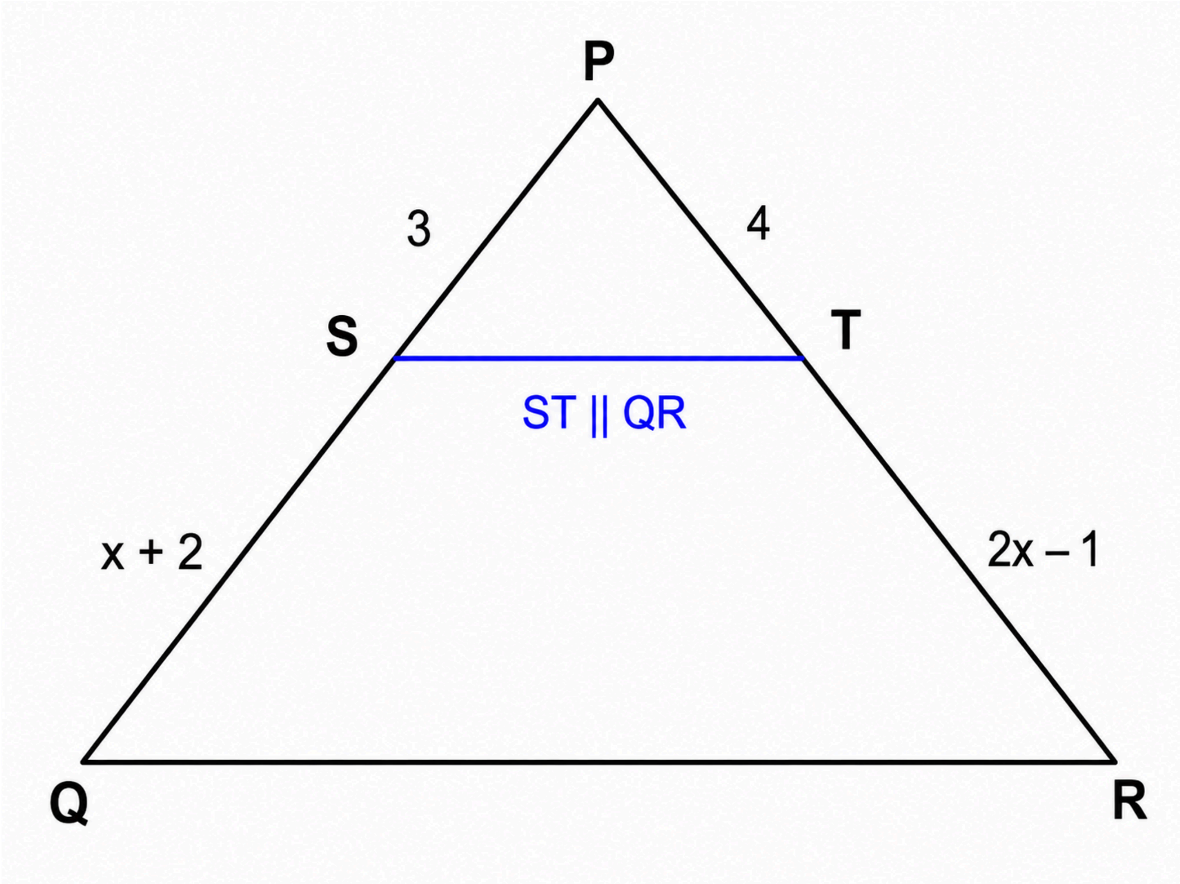
Similarity and area scaling

Answer **C**

QUESTION 21

In triangle PQR, point S lies on PQ and point T lies on PR. The line ST is parallel to QR.

The lengths are labelled as follows: $PS = 3$ cm, $SQ = (x + 2)$ cm, $PT = 4$ cm, $TR = (2x - 1)$ cm.



Given that the area of triangle PST is $12\ \mathrm{cm}^2$, what is the area of trapezium $STQR$?

A $96\ \mathrm{cm}^2$

B $120\ \mathrm{cm}^2$

C $135\ \mathrm{cm}^2$

D $144\ \mathrm{cm}^2$

E $147\ \mathrm{cm}^2$

WORKED SOLUTION

Since $ST \parallel QR$, triangles PST and PQR are similar, so

$$\frac{PS}{PQ} = \frac{PT}{PR}.$$

Now $PQ = 3 + (x + 2) = x + 5$ and $PR = 4 + (2x - 1) = 2x + 3$. Therefore

$$\frac{3}{x + 5} = \frac{4}{2x + 3}.$$

So $6x + 9 = 4x + 20$, giving $x = \frac{11}{2}$.

Then

$$\frac{PS}{PQ} = \frac{3}{21/2} = \frac{2}{7}.$$

Hence the area scale factor is $\left(\frac{2}{7}\right)^2 = \frac{4}{49}$.

So if area $PST = 12$, then area $PQR = 12 \cdot \frac{49}{4} = 147$.

Thus the area of trapezium $STQR$ is $147 - 12 = \boxed{135\text{ cm}^2}$.

M1-22

Probability with two colours

Answer **B**

QUESTION 22

Charlie has a bowl containing red sweets and green sweets only. There are 9 more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g. Then the number of red sweets is $g+9$, so total sweets are $2g+9$.

The probability condition is

$$\frac{\binom{g+9}{2}+\binom{g}{2}}{\binom{2g+9}{2}}=\frac{4}{7}.$$

This simplifies to

$$(g-6)(g+18)=0.$$

So $g=6$. Then red sweets $=15$, giving total sweets $=21$.

M1-23

Successive translations and vector equations

Answer **D**

QUESTION 23

The point $(-1,5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2-q \end{pmatrix}$, and the second is by $\begin{pmatrix} q+1 \\ p-3 \end{pmatrix}$.

The final image is $(8,4)$.

What is the value of p^2+q^2?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p+q+1 \\ p-q-1 \end{pmatrix}.$$

Since $(-1, 5) \to (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p+q=8, \qquad p-q=0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

M1-24 Quadratic graphs and minimum values Answer A

QUESTION 24

Consider the family of graphs

$$y = ax^2 + 2(a-1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x-axis?

A $a > \frac{1}{5}$

B $a > 0$

C $a \geq \frac{1}{5}$

D $a < 0$

E $0 < a < \frac{1}{5}$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a - 1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a-1))^2}{4a} = a + 3 - \frac{(a-1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

M1-25 Counting constrained selections Answer **E**

QUESTION 25

An online company sells storage containers. The following items are available:

Capacity of container	Number available
2 litres	2
3 litres	3

Capacity of container	Number available
5 litres	4

A customer chooses a selection of containers with total capacity 23 litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a, b, c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \leq a \leq 2, 0 \leq b \leq 3, 0 \leq c \leq 4$.

The valid selections are

- $(0, 1, 4)$
- $(1, 2, 3)$
- $(2, 3, 2)$

Two of the three contain at least one 2-litre container, so the probability is $\boxed{\frac{2}{3}}$.

M1-26

Similarity and three-dimensional scaling

Answer C

QUESTION 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13}\text{ cm}}.$$

M1-27

Relative speed in a race

Answer B

QUESTION 27

Alex, Cameron and Sam are running a $400\ \mathrm{m}$ race. They each run at a different constant speed.

When Alex finishes, Cameron has run $300\ \mathrm{m}$ and Sam has run $240\ \mathrm{m}$.

Alex beats Sam by $33\frac{1}{3}$ seconds.

What is Cameron's speed?

- **A** $5\ \mathrm{m\ s^{-1}}$
- **B** $6\ \mathrm{m\ s^{-1}}$
- **C** $6.4\ \mathrm{m\ s^{-1}}$
- **D** $7\ \mathrm{m\ s^{-1}}$
- **E** $8\ \mathrm{m\ s^{-1}}$

WORKED SOLUTION

Let Alex's speed be A, Cameron's speed be C, and Sam's speed be S.

When Alex finishes, the elapsed time is $\frac{400}{A}$. In that time, Cameron runs 300 m, so

$$C \cdot \frac{400}{A} = 300 \Rightarrow \frac{C}{A} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$S \cdot \frac{400}{A} = 240 \Rightarrow \frac{S}{A} = \frac{3}{5}.$$

Thus $S = \frac{3}{5}A$.

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8\ \mathrm{m\ s^{-1}}$. Hence

$$C = \frac{3}{4}A = 6\ \mathrm{m\ s^{-1}}.$$

Therefore Cameron's speed is $\boxed{6\ \mathrm{m\ s^{-1}}}$.

LEVEL 2 · HARDER THAN ESAT

Physics

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Charage, current, power and heating

Answer D

QUESTION 1

A resistor has a constant potential difference of $9.0\ \mathrm{V}$ across it. A total charge of $180\ \mathrm{C}$ passes through it in $4.00\ \mathrm{min}$.

The resistor is then left connected for a total of $6.00\ \mathrm{min}$. Assume all the energy dissipated warms a $0.250\ \mathrm{kg}$ metal block of specific heat capacity $400\ \mathrm{J\ kg^{-1} K^{-1}}$, and that there are no losses.

What is the temperature rise of the block?

- **A** $12.0\ \mathrm{K}$
- **B** $18.0\ \mathrm{K}$
- **C** $21.6\ \mathrm{K}$
- **D** $24.3\ \mathrm{K}$
- **E** $27.0\ \mathrm{K}$

WORKED SOLUTION

The current is

$$I = \frac{Q}{t} = \frac{180}{240} = 0.75\ \mathrm{A}.$$

So the power is

$$P = VI = 9.0 \times 0.75 = 6.75\ \mathrm{W}.$$

Over $6.00\ \mathrm{min} = 360\ \mathrm{s}$, the energy transferred is

$$E = Pt = 6.75 \times 360 = 2430\ \mathrm{J}.$$

Hence

$$\Delta T = \frac{E}{mc} = \frac{2430}{0.250 \times 400} = 24.3\ \mathrm{K}.$$

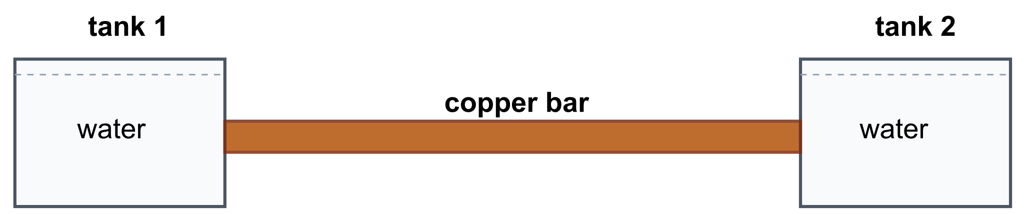
P-02

Thermal conduction and scaling

Answer **D**

QUESTION 2

Two large copper tanks of water are connected by a solid cylindrical copper bar. Initially the water temperatures are $20^{\circ}\mathrm{C}$ and $80^{\circ}\mathrm{C}$, and the bar has length L and diameter d.



The system is modified so that the temperatures become $35^{\circ}\mathrm{C}$ and $95^{\circ}\mathrm{C}$, the bar length becomes $0.75L$, and the diameter becomes $1.5d$.

By what factor does the rate of conduction of thermal energy along the bar change?

A 0.75

B 1.5

C 2.25

D 3.0

E 4.0

WORKED SOLUTION

For conduction,

$$P \propto \frac{A\Delta T}{L}.$$

The temperature difference stays $60^\circ C$, so ΔT is unchanged. The area factor is

$$\left(\frac{1.5d}{d}\right)^2 = 2.25.$$

The length factor is

$$\frac{L}{0.75L} = \frac{4}{3}.$$

Hence the overall factor is

$$2.25 \times \frac{4}{3} = 3.0.$$

P-03

Forces, towing and tension

Answer C

QUESTION 3

A car of mass 1400 kg tows a caravan of mass 1000 kg along a horizontal road at constant speed. The driving force from the car's engine is then increased by 3000 N.

At one instant during the resulting acceleration, the resistive force on the car has increased by 200 N and the resistive force on the caravan has increased by 400 N, compared with their values at constant speed.

By how much has the tension in the tow bar increased compared with its value at constant speed?

A $600\ \mathrm{N}$

B $1000\ \mathrm{N}$

C $1400\ \mathrm{N}$

D $1600\ \mathrm{N}$

E $1800\ \mathrm{N}$

WORKED SOLUTION

For the combined car–caravan system, the increase in the forward resultant is

$$3000 - 200 - 400 = 2400\ \mathrm{N}.$$

The total mass is $2400\ \mathrm{kg}$, so the acceleration is

$$a = \frac{2400}{2400} = 1.0\ \mathrm{m\ s^{-2}}.$$

Let T_0 be the original tow-bar tension and R_0 the original resistive force on the caravan. At constant speed, $T_0 = R_0$.

During the acceleration, let the new tension be T_1. The caravan equation is

$$T_1 - (R_0 + 400) = 1000(1.0).$$

Hence $T_1 - R_0 = 1400\ \mathrm{N}$. Since $T_0 = R_0$,

$$T_1 - T_0 = \boxed{1400\ \mathrm{N}}.$$

P-04

Impulse and pressure

Answer C

QUESTION 4

A small piece of space debris of mass $0.10\ \mathrm{kg}$ strikes the International Space Station at a relative speed of $1.5\times 10^{4}\ \mathrm{m\ s^{-1}}$. The debris comes to rest relative to the station in $0.010\ \mathrm{s}$.

Assume the average contact area during the impact is $2.0\ \mathrm{cm}^2$.

What is the average pressure exerted on the station during the impact?

- **A** $7.5\times 10^{6}\ \mathrm{Pa}$
- **B** $7.5\times 10^{7}\ \mathrm{Pa}$
- **C** $7.5\times 10^{8}\ \mathrm{Pa}$
- **D** $1.5\times 10^{8}\ \mathrm{Pa}$
- **E** $1.5\times 10^{9}\ \mathrm{Pa}$

WORKED SOLUTION

The average force magnitude is

$$F = \frac{\Delta p}{\Delta t} = \frac{0.10\times 1.5\times 10^{4}}{0.010} = 1.5\times 10^{5}\ \mathrm{N}.$$

The contact area is

$$2.0\ \mathrm{cm}^2 = 2.0\times 10^{-4}\ \mathrm{m}^2.$$

So the pressure is

$$p = \frac{F}{A} = \frac{1.5\times 10^{5}}{2.0\times 10^{-4}} = 7.5\times 10^{8}\ \mathrm{Pa}.$$

P-05

Light, Doppler shift and photon energy

Answer B

QUESTION 5

A star is moving away from a space telescope. The star emits light of wavelength $500\ \mathrm{nm}$, but the telescope detects it with wavelength $502\ \mathrm{nm}$.

What is the percentage decrease in the detected frequency, and hence in the energy of each detected photon?

A 0.20%

B 0.40%

C 0.80%

D 1.0%

E 2.0%

WORKED SOLUTION

In vacuum, $c = f\lambda$, so $f \propto \dfrac{1}{\lambda}$.

The fractional decrease in frequency is therefore approximately the same as the fractional increase in wavelength:

$$\frac{502 - 500}{502} \approx \frac{2}{502} \approx 0.00398.$$

This is about 0.40%. Since photon energy is $E = hf$, the percentage decrease in photon energy is the same.

P-06

Efficiency, power and useful work

Answer D

QUESTION 6

An electric motor is supplied with power at $0.800\ \mathrm{kW}$. Its efficiency is 60%, and it runs for half an hour while lifting a load vertically at constant speed.

Ignoring any change in the motor itself, what maximum mass can be lifted through a height of $180\ \mathrm{m}$? Take $g = 10\ \mathrm{N\ kg^{-1}}$.

- **A** $120\ \mathrm{kg}$
- **B** $240\ \mathrm{kg}$
- **C** $360\ \mathrm{kg}$
- **D** $480\ \mathrm{kg}$
- **E** $720\ \mathrm{kg}$

WORKED SOLUTION

Input power is $800\ \mathrm{W}$, so useful power is

$$0.60 \times 800 = 480\ \mathrm{W}.$$

Over $1800\ \mathrm{s}$, the useful energy is

$$E = 480 \times 1800 = 864000\ \mathrm{J}.$$

This equals the gain in gravitational potential energy:

$$mgh = 864000.$$

Hence

$$m = \frac{864000}{10 \times 180} = 480\ \mathrm{kg}.$$

P-07

Ideal gases and density

Answer **D**

QUESTION 7

Air is trapped in a cylinder by a freely moving piston. Initially the trapped air has density ρ at temperature $300\ \mathrm{K}$.

The external load on the piston is adjusted so that the pressure of the trapped air becomes 20% greater than its initial pressure. The load is then kept unchanged, so the pressure remains at this value while the trapped air cools to $270\ \mathrm{K}$. No gas escapes.

Assume the air behaves as an ideal gas. What is the final density of the trapped air?

A 1.09ρ

B 1.20ρ

C 1.25ρ

D $\frac{4}{3}\rho$

E 1.44ρ

WORKED SOLUTION

For a fixed mass of an ideal gas,

$$p = \rho R T \quad \Rightarrow \quad \rho \propto \frac{p}{T}.$$

The final pressure is 1.20 times the initial pressure, while the temperature changes from $300\ \mathrm{K}$ to $270\ \mathrm{K}$. Therefore

$$\frac{\rho_f}{\rho_i} = \frac{1.20}{270/300} = 1.20 \times \frac{300}{270} = \frac{4}{3}.$$

Hence $\boxed{\rho_f = \frac{4}{3}\rho}$.

P-08

Current, magnetic force and geometry

Answer C

QUESTION 8

A steady current flows in a conducting wire. A charge of $48\ \mathrm{C}$ passes a point on the wire in $2.0\ \mathrm{min}$.

A straight $36\ \mathrm{cm}$ section of the wire lies in a uniform magnetic field of strength $0.20\ \mathrm{T}$. This section makes an angle of 30° to the direction of the field.

What is the magnitude of the magnetic force on this section?

A $2.88 \times 10^{-3}\ \mathrm{N}$

B $7.2 \times 10^{-3}\ \mathrm{N}$

C $1.44 \times 10^{-2}\ \mathrm{N}$

D $2.88 \times 10^{-2}\ \mathrm{N}$

E $5.76 \times 10^{-2}\ \mathrm{N}$

WORKED SOLUTION

The current is

$$I = \frac{Q}{t} = \frac{48}{120} = 0.40\ \mathrm{A}.$$

The magnetic force on a wire is

$$F = BIL\sin\theta.$$

So

$F = 0.20 \times 0.40 \times 0.36 \times \sin 30^{\circ} = 0.20 \times 0.40 \times 0.36 \times 0.5 = 1.44 \times 10^{-2}$

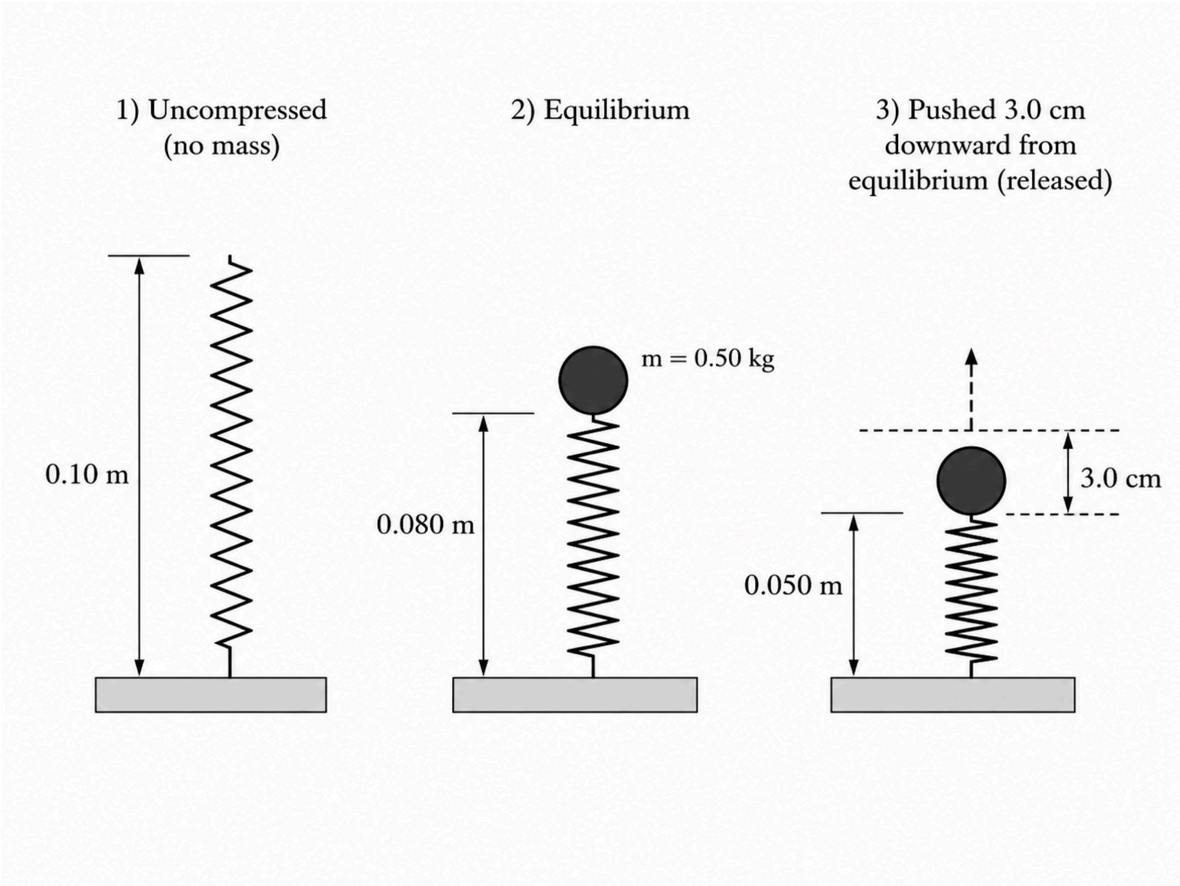
P-09

Springs, equilibrium and oscillation energy

Answer **D**

QUESTION 9

A light spring has uncompressed length 0.10 m. When an object of mass 0.50 kg rests on top of the spring in equilibrium, the spring length is 0.080 m.



The mass is then pushed a further 3.0 cm downward from equilibrium and released, moving on a frictionless vertical guide.

What is the speed of the mass as it passes back through its equilibrium position?

Take $g = 10 \text{ N kg}^{-1}$.

A 0.30 m s^{-1}

B $0.47\ \mathrm{m\ s^{-1}}$

C $0.60\ \mathrm{m\ s^{-1}}$

D $0.67\ \mathrm{m\ s^{-1}}$

E $0.95\ \mathrm{m\ s^{-1}}$

WORKED SOLUTION

At equilibrium, the spring is compressed by $0.020\ \mathrm{m}$, so

$$kx = mg \quad \Rightarrow \quad k = \frac{0.50 \times 10}{0.020} = 250\ \mathrm{N\ m^{-1}}.$$

If the mass is displaced by an additional $a = 0.030\ \mathrm{m}$ from equilibrium, then at the equilibrium position all of the extra elastic/gravitational energy has become kinetic energy:

$$\frac{1}{2}ka^2 = \frac{1}{2}mv^2.$$

Hence

$$v = a\sqrt{\frac{k}{m}} = 0.030\sqrt{\frac{250}{0.50}} = 0.030\sqrt{500} \approx 0.67\ \mathrm{m\ s^{-1}}.$$

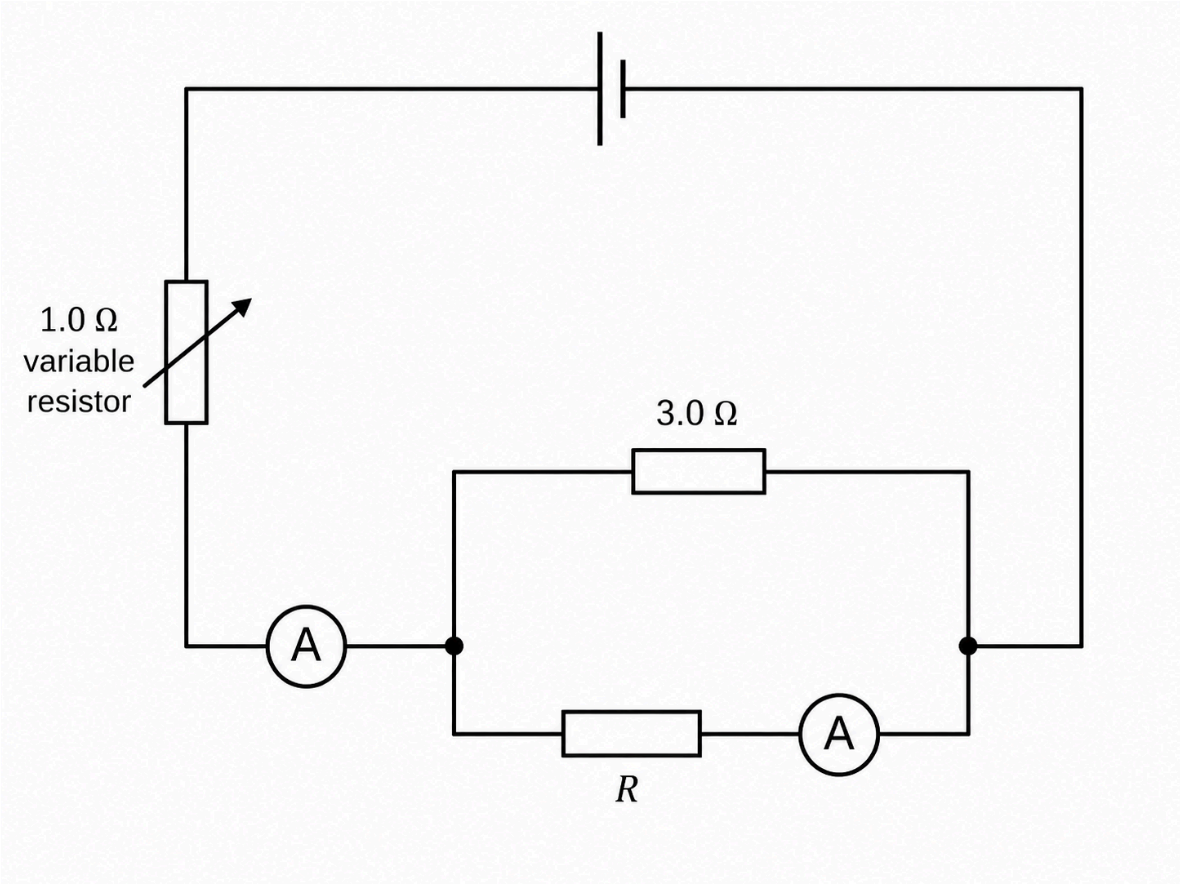
P-10

Circuits, current splitting and emf

Answer C

QUESTION 10

The circuit includes a variable resistor in series with a parallel network containing a $3.0\ \Omega$ resistor and a resistor R. The two ammeter readings are shown.



The variable resistor has resistance $1.0\,\Omega$. Assume ideal ammeters and negligible internal resistance of the supply.

What is the emf of the supply?

A 4.5 V

B 6.6 V

C 7.8 V

D 9.0 V

E 12.3 V

WORKED SOLUTION

The right ammeter is in the R branch, so the current in R is 1.8 A.

The current in the $3.0\,\Omega$ branch is

$$3.3 - 1.8 = 1.5\ \mathrm{A}.$$

So the p.d. across the parallel network is

$$V_{\parallel} = 1.5 \times 3.0 = 4.5\ \mathrm{V}.$$

The current through the series variable resistor is the total current $3.3\ \mathrm{A}$, so its p.d. is

$$V_{\mathrm{var}} = 3.3 \times 1.0 = 3.3\ \mathrm{V}.$$

Hence the supply emf is

$$E = 4.5 + 3.3 = 7.8\ \mathrm{V}.$$

P-11

Nuclides and algebraic reasoning

Answer **E**

QUESTION 11

Element Q has several isotopes. Consider the two isotopes

$$^{3x-7}_{x}Q \qquad \text{and} \qquad ^{2x+3}_{x}Q.$$

The first contains 6 more neutrons than the second.

Another isotope of Q is

$$^{3x+1}_{x}Q.$$

How many neutrons does this third isotope contain?

A 19

B 21

C 25

D 29

E 33

WORKED SOLUTION

The neutron numbers are:

first isotope: $(3x - 7) - x = 2x - 7$

second isotope: $(2x + 3) - x = x + 3$

Given that the first has 6 more neutrons,

$$(2x - 7) - (x + 3) = 6.$$

So

$$x - 10 = 6 \Rightarrow x = 16.$$

For the third isotope, the number of neutrons is

$$(3x + 1) - x = 2x + 1 = 2(16) + 1 = 33.$$

P-12 Momentum graphs, friction and power Answer **D**

QUESTION 12

A motorised trolley of mass 2.4×10^3 kg accelerates from rest along a horizontal track. The friction force is constant at 1.8×10^3 N.

The graph shows how the momentum of the trolley varies with time.

What is the power delivered by the trolley's motor at $t = 4.0$ s?

A 1.5×10^2 W

B 3.0×10^2 W

C 6.3×10^2 W

D 1.05×10^3 W

E 2.1×10^3 W

WORKED SOLUTION

The gradient of a momentum–time graph gives the resultant force. From the graph, the momentum reaches 12×10^2 kg m s^{-1} at 4.0 s, so

$$F_{net} = \frac{12 \times 10^2}{4} = 3.0 \times 10^2 \text{ N}.$$

The motor must provide both the resultant force and the force needed to overcome friction:

$$F_{motor} = 3.0 \times 10^2 + 1.8 \times 10^3 = 2.1 \times 10^3 \text{ N}.$$

At $t = 4.0$ s,

$$v = \frac{p}{m} = \frac{1.2 \times 10^3}{2.4 \times 10^3} = 0.50 \text{ m s}^{-1}.$$

Therefore the power delivered is

$$P = F_{motor} v = (2.1 \times 10^3)(0.50) = \boxed{1.05 \times 10^3 \text{ W}}.$$

P-13

Work–energy and resistive forces

Answer C

QUESTION 13

A block of mass 6.0 kg is pushed along a rough horizontal surface by a constant force of 8.0 N. It accelerates uniformly from rest and reaches speed 2.0 m s^{-1} after 4.0 s.

What fraction of the work done by the pushing force during this 4.0 s is converted into thermal energy against resistive forces?

A $\frac{3}{8}$

B $\frac{1}{2}$

C $\frac{5}{8}$

D $\frac{3}{4}$

E $\frac{5}{6}$

WORKED SOLUTION

The acceleration is

$$a = \frac{2.0}{4.0} = 0.50 \text{ m s}^{-2}.$$

The resultant force is therefore

$$F_{\text{net}} = ma = 6.0 \times 0.50 = 3.0 \text{ N}.$$

So the resistive force is

$$8.0 - 3.0 = 5.0 \text{ N}.$$

The displacement in 4.0 s is

$$s = \frac{1}{2}at^2 = \frac{1}{2}(0.50)(4.0)^2 = 4.0 \text{ m}.$$

Work done by the pushing force:

$$W_{\text{push}} = 8.0 \times 4.0 = 32 \text{ J}.$$

Thermal energy against resistance:

$$W_{\text{res}} = 5.0 \times 4.0 = 20 \text{ J}.$$

Hence the fraction is

$$\frac{20}{32} = \frac{5}{8}.$$

P-14

Specific heat capacity and gradients

Answer C

QUESTION 14

A metal block of mass M is heated at a constant rate P. The graph of temperature rise ΔT against time t is a straight line of gradient k.

A second block is made of the same material, has mass $3M$, and is heated at constant rate $2P$.

What is the gradient of its ΔT-against-t graph?

A $\frac{k}{6}$

B $\frac{k}{3}$

C $\frac{2k}{3}$

D $\frac{3k}{2}$

E $6k$

WORKED SOLUTION

For the first block,

$$P = Mc\frac{dT}{dt}.$$

Hence

$$k = \frac{P}{Mc}.$$

For the second block, the gradient is

$$k_2 = \frac{2P}{3Mc} = \frac{2}{3}\frac{P}{Mc} = \frac{2k}{3}.$$

P-15

Transformers, efficiency and heating

Answer D

QUESTION 15

A non-ideal transformer has 100 turns on the primary coil and 25 turns on the secondary coil. It is supplied with $3.0\ \mathrm{kW}$ of electrical power at a current of $12.5\ \mathrm{A}$.

The output voltage is the same as for an ideal transformer, but the current in the secondary coil is $40\ \mathrm{A}$.

The transformer runs for $15\ \mathrm{min}$. Assume all power lost by inefficiency remains in the transformer, whose thermal capacity is $2.0\times\ 10^4\ \mathrm{J\ K^{-1}}$.

What is the temperature rise of the transformer?

- **A** $12\ \mathrm{K}$
- **B** $18\ \mathrm{K}$
- **C** $24\ \mathrm{K}$
- **D** $27\ \mathrm{K}$
- **E** $30\ \mathrm{K}$

WORKED SOLUTION

Input voltage is

$$V_p = \frac{P_{in}}{I_p} = \frac{3000}{12.5} = 240\ \mathrm{V}.$$

For the turns ratio,

$$\frac{V_s}{V_p} = \frac{25}{100} = \frac{1}{4},$$

so $V_s = 60\ \mathrm{V}$.

Thus the output power is

$$P_{out} = V_s I_s = 60 \times 40 = 2400\ \mathrm{W}.$$

Power lost is

$$P_{loss} = 3000 - 2400 = 600\ \mathrm{W}.$$

In $900\ \mathrm{s}$, the thermal energy gained is

$$E = 600 \times 900 = 5.4 \times 10^5\ \mathrm{J}.$$

So the temperature rise is

$$\Delta T = \frac{E}{C} = \frac{5.4 \times 10^5}{2.0 \times 10^4} = 27\ \mathrm{K}.$$

P-16

Series circuits and power scaling

Answer **D**

QUESTION 16

A set of decorative lights consists of 20 identical lamps connected in series to a d.c. supply of constant voltage. The total power transferred by all 20 lamps together is P.

One lamp fails so that it becomes short-circuited and has zero resistance. The remaining 19 lamps are still lit.

By what factor does the power transferred to *each remaining lamp* increase?

A $\dfrac{19}{20}$

B $\dfrac{20}{19}$

C $\frac{361}{400}$

D $\frac{400}{361}$

E $\frac{400}{19}$

WORKED SOLUTION

If each lamp has resistance R, then the original total resistance is $20R$, so current is

$$I = \frac{V}{20R}.$$

After one lamp is short-circuited, the total resistance is $19R$, so current becomes

$$I' = \frac{V}{19R}.$$

Power in each functioning lamp is proportional to I^2R. Therefore the factor increase is

$$\left(\frac{I'}{I}\right)^2 = \left(\frac{V/(19R)}{V/(20R)}\right)^2 = \left(\frac{20}{19}\right)^2 = \frac{400}{361}.$$

P-17 Electromagnetic spectrum Answer E

QUESTION 17

The wavelength range of visible light is about $400\ \mathrm{nm}$ to $700\ \mathrm{nm}$. Light of frequency $6.0 \times 10^{14}\ \mathrm{Hz}$ is green. Microwaves used in cooking have wavelength $12\ \mathrm{cm}$.

Which statements are correct?

1. Light of frequency $7.5 \times 10^{14}\ \mathrm{Hz}$ is red.

2. Microwaves used in cooking have frequency 2.5×10^{9} Hz.

3. Electromagnetic radiation of frequency 3.0×10^{13} Hz can be used for thermal imaging of buildings.

Take $c = 3.0 \times 10^{8}\ \mathrm{m\ s^{-1}}$.

A none of them

B 1 only

C 2 only

D 3 only

E 2 and 3 only

WORKED SOLUTION

Statement 1 is false: a higher visible frequency corresponds to bluer/violet light, not red.

For statement 2,

$$f = \frac{c}{\lambda} = \frac{3.0 \times 10^{8}}{0.12} = 2.5 \times 10^{9}\ \mathrm{Hz},$$

so statement 2 is true.

Statement 3 is also true: 3.0×10^{13} Hz lies in the infrared region, which is used in thermal imaging.

P-18 Series circuits and filament lamps Answer **D**

QUESTION 18

A $6.0\ \mathrm{V}$ battery is connected in series with an $8.0\,\Omega$ resistor, a filament lamp and an ammeter. The ammeter reading is $0.25\ \mathrm{A}$.

Which pair gives the potential difference across the lamp and the resistance of the lamp at this operating point?

- **A** $2.0\ \mathrm{V},\ 8.0\,\Omega$
- **B** $4.0\ \mathrm{V},\ 8.0\,\Omega$
- **C** $4.0\ \mathrm{V},\ 16\,\Omega$ and power $0.50\ \mathrm{W}$
- **D** $4.0\ \mathrm{V},\ 16\,\Omega$ and power $1.0\ \mathrm{W}$
- **E** $6.0\ \mathrm{V},\ 24\,\Omega$

WORKED SOLUTION

The resistor has p.d.

$$V_R = IR = 0.25 \times 8.0 = 2.0\ \mathrm{V}.$$

So the lamp has

$$V_L = 6.0 - 2.0 = 4.0\ \mathrm{V}.$$

Its resistance at this operating point is

$$R_L = \frac{V_L}{I} = \frac{4.0}{0.25} = 16\,\Omega.$$

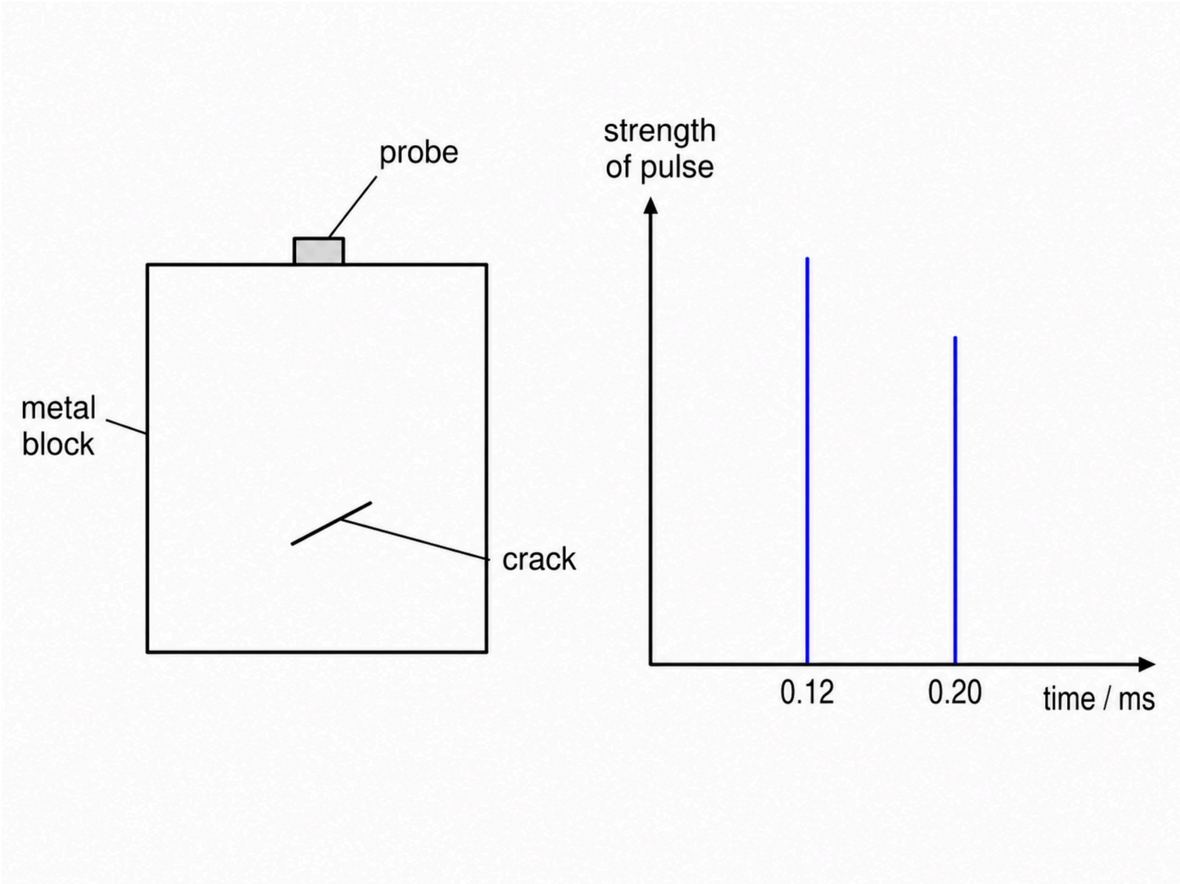
Its power is

$$P_L = V_L I = 4.0 \times 0.25 = 1.0\ \mathrm{W}.$$

P-19 Ultrasound and echo timing Answer **C**

QUESTION 19

Ultrasound is used to find a crack inside a cuboid metal block. A probe is held against the top surface and sends a short pulse into the block. Reflections return from the crack and from the bottom surface.



The reflections return after $0.12\ \mathrm{ms}$ and $0.20\ \mathrm{ms}$. The speed of ultrasound in the metal is $5000\ \mathrm{m\ s^{-1}}$.

What is the ratio

$$\frac{\text{thickness of block}}{\text{depth of crack below the top surface}}?$$

A $2 : 1$

B $3 : 2$

C $5 : 3$

D $4 : 3$

E $5 : 4$

WORKED SOLUTION

Echo times are round-trip times, so

depth of crack:

$$d = \frac{vt}{2} = \frac{5000 \times 0.12 \times 10^{-3}}{2} = 0.30 \text{ m}.$$

Thickness of block:

$$h = \frac{5000 \times 0.20 \times 10^{-3}}{2} = 0.50 \text{ m}.$$

Therefore

$$h : d = 0.50 : 0.30 = 5 : 3.$$

P-20 Kinematics and minimum-time travel **Answer C**

QUESTION 20

A straight railway line between two cities is 60 km long. A train starts from rest, accelerates uniformly at 1.5 m s^{-2} to a maximum speed of 120 m s^{-1}, then later brakes uniformly at 2.0 m s^{-2} to come to rest at the second city.

What is the minimum possible journey time?

A 430 s

B 500 s

C 570 s

D 640 s

E 860 s

WORKED SOLUTION

Acceleration phase:

$$t_1 = \frac{120}{1.5} = 80 \text{ s}, \qquad s_1 = \frac{120^2}{2(1.5)} = 4800 \text{ m}.$$

Braking phase:

$$t_2 = \frac{120}{2.0} = 60 \text{ s}, \qquad s_2 = \frac{120^2}{2(2.0)} = 3600 \text{ m}.$$

Total distance used in speeding up and slowing down:

$$4800 + 3600 = 8400 \text{ m}.$$

Remaining cruising distance:

$$60000 - 8400 = 51600 \text{ m}.$$

Cruising time:

$$t_3 = \frac{51600}{120} = 430 \text{ s}.$$

Total time:

$$80 + 60 + 430 = 570 \text{ s}.$$

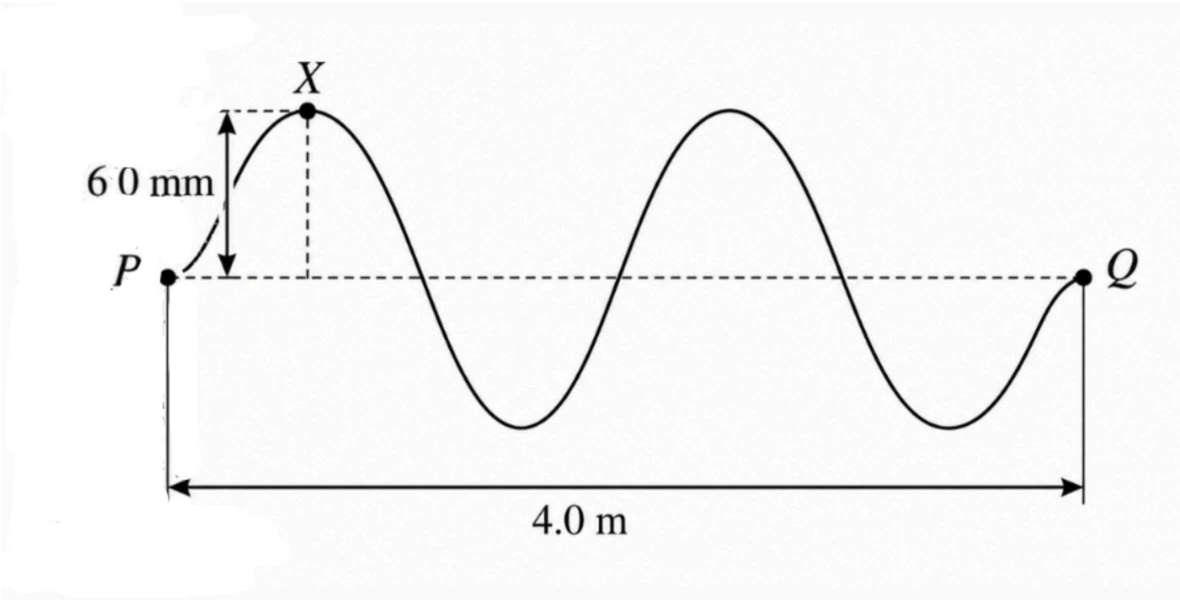
P-21

Waves, phase and SHM

Answer B

QUESTION 21

A transverse wave on a string has speed $500 \mathrm{~m~s^{-1}}$. The horizontal distance between points P and Q shown is $4.0 \mathrm{~m}$.



At time $\mathrm{t} = 0$, point X is at its maximum displacement of $6.0 \mathrm{~mm}$ above equilibrium.

What is the displacement of point X at $\mathrm{t} = 7.0 \mathrm{~ms}$, and in which direction is it moving at that instant?

- **A** $6.0 \mathrm{~mm}$ above equilibrium, moving downward
- **B** $0 \mathrm{~mm}$, moving upward
- **C** $0 \mathrm{~mm}$, moving downward
- **D** $6.0 \mathrm{~mm}$ below equilibrium, moving upward
- **E** between 0 and $6.0 \mathrm{~mm}$ below equilibrium, moving downward

WORKED SOLUTION

From the diagram, the distance from P to Q spans two complete wavelengths. Therefore

$$2\lambda = 4.0\ \mathrm{m},$$

so

$$\lambda = 2.0\ \mathrm{m}.$$

The frequency of the wave is therefore

$$f = \frac{v}{\lambda} = \frac{500}{2.0} = 250\ \mathrm{Hz}.$$

Hence the period is

$$T = \frac{1}{f} = \frac{1}{250} = 0.004\ \mathrm{s} = 4.0\ \mathrm{ms}.$$

Now

$$7.0\ \mathrm{ms} = 1.75T.$$

This is equivalent to $0.75T$ after one complete cycle. Starting from maximum positive displacement, after $0.75T$ the point is at equilibrium and moving upward. Therefore the displacement is

$$\boxed{0\ \mathrm{mm,\ moving\ upward}}.$$

P-22

Radioactivity and half-life

Answer **B**

QUESTION 22

A radioactive nuclide X decays in a single stage to a stable nuclide R. Another radioactive nuclide Y decays in a single stage to a stable nuclide S.

When a rock formed, it contained equal numbers of atoms of X, Y, R and S.

The half-life of X is T years and the half-life of Y is $2T$ years.

What is the value of

$\dfrac{\text{number of atoms of R}}{\text{number of atoms of S}}$ at a time $4T$ years after the rock formed?

A $\dfrac{17}{20}$

B $\dfrac{31}{28}$

C $\dfrac{5}{4}$

D $\dfrac{6}{5}$

E 2

WORKED SOLUTION

Let each initial number be N.

For X: after $4T$, four half-lives have passed, so

$$X = \frac{N}{16}.$$

Thus $\frac{15N}{16}$ atoms have decayed into R, giving

$$R = N + \frac{15N}{16} = \frac{31N}{16}.$$

For Y: after $4T$, two half-lives have passed, so

$$Y = \frac{N}{4}.$$

Hence $\frac{3N}{4}$ atoms have decayed into S, giving

$$S = N + \frac{3N}{4} = \frac{7N}{4}.$$

Therefore

$$\frac{R}{S} = \frac{31N/16}{7N/4} = \frac{31}{28}.$$

P-23

Waves and relative motion

Answer D

QUESTION 23

A ship travels into a wave that is moving in the opposite direction to the ship. The ship has speed $8.0\ \mathrm{m\ s^{-1}}$ and the wave has speed $3.0\ \mathrm{m\ s^{-1}}$.

The front of the ship rises and falls with a time period of $8.0\ \mathrm{s}$.

What are the wavelength of the wave and the period of the wave for a stationary buoy?

- **A** $24\ \mathrm{m},\ 8.0\ \mathrm{s}$
- **B** $40\ \mathrm{m},\ 13.3\ \mathrm{s}$
- **C** $64\ \mathrm{m},\ 21.3\ \mathrm{s}$
- **D** $88\ \mathrm{m},\ 29.3\ \mathrm{s}$
- **E** $88\ \mathrm{m},\ 8.0\ \mathrm{s}$

WORKED SOLUTION

Relative speed between ship and wave crests is

$$8.0 + 3.0 = 11.0\ \mathrm{m\ s^{-1}}.$$

So the wavelength is

$$\lambda = v_{rel}T_{ship} = 11.0 \times 8.0 = 88\ \mathrm{m}.$$

For a stationary buoy, the wave period is

$$T = \frac{\lambda}{v_{wave}} = \frac{88}{3.0} = 29.3 \text{ s.}$$

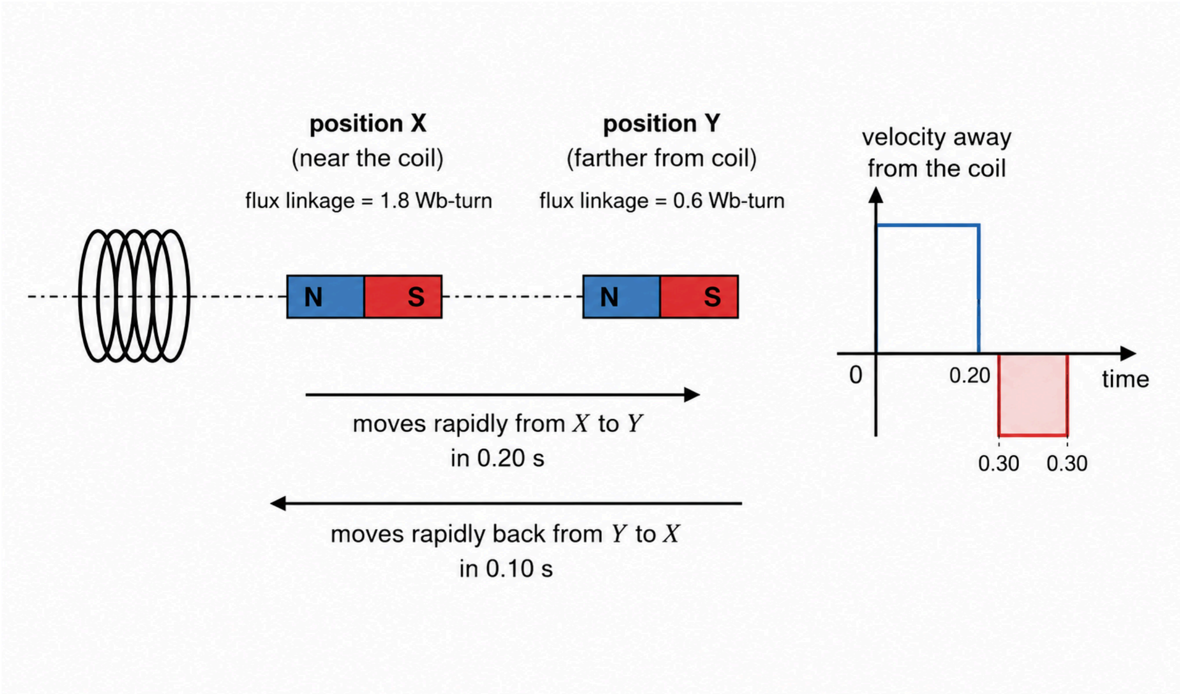
P-24

Electromagnetic induction

Answer **D**

QUESTION 24

A bar magnet is placed at position X close to one end of a coil and on its axis. It is moved rapidly to a more distant position Y, and then back to X.



The flux linkage of the coil is $1.8\ \mathrm{Wb\text{-}turn}$ when the magnet is at X and $0.6\ \mathrm{Wb\text{-}turn}$ when it is at Y. The motion from X to Y takes 0.20 s, while the return motion from Y to X takes 0.10 s.

What is the greatest magnitude of the induced e.m.f. during the complete motion?

A 3.0 V

B 6.0 V

C 9.0 V

D 12 V

E 18 V

WORKED SOLUTION

The change in flux linkage between X and Y is

$$\Delta(N\Phi) = 1.8 - 0.6 = 1.2 \text{ Wb-turn.}$$

For the outward motion, the average induced e.m.f. magnitude is

$$E_1 = \frac{1.2}{0.20} = 6.0 \text{ V.}$$

For the return motion, the same flux-linkage change happens in 0.10 s, so

$$E_2 = \frac{1.2}{0.10} = 12 \text{ V.}$$

The greatest magnitude is 12 V.

P-25 Seismic waves Answer B

QUESTION 25

When an earthquake occurs, both P-waves and S-waves are produced at the same time and travel along the same path. P-waves travel at 5.0 km s^{-1} and S-waves travel at 3.0 km s^{-1}.

A seismic monitoring station detects the P-waves 30 s before it detects the S-waves.

How long after the earthquake does the P-wave reach the station?

A 30 s

B 45 s

C $60\ \mathrm{s}$

D $75\ \mathrm{s}$

E $90\ \mathrm{s}$

WORKED SOLUTION

Let the distance to the station be d km. Then

$$\frac{d}{3.0} - \frac{d}{5.0} = 30.$$

So

$$d\left(\frac{1}{3} - \frac{1}{5}\right) = 30 \Rightarrow d\left(\frac{2}{15}\right) = 30 \Rightarrow d = 225\ \mathrm{km}.$$

The P-wave travel time is therefore

$$t_P = \frac{225}{5.0} = 45\ \mathrm{s}.$$

P-26

Pressure, ideal gas law and geometry

Answer **C**

QUESTION 26

A tall smooth cylinder contains air at atmospheric pressure $1.00 \times 10^5\ \mathrm{Pa}$. The density of the air is $1.20\ \mathrm{kg\ m^{-3}}$. A heavy piston of mass $50.0\ \mathrm{kg}$ and cross-sectional area $0.0200\ \mathrm{m^2}$ is placed on top and allowed to fall slowly until it rests in equilibrium.

The initial height of the trapped air column is $0.50\ \mathrm{m}$. Assume the air behaves as an ideal gas and its temperature remains constant. Take $g = 10\ \mathrm{N\ kg^{-1}}$.

By how far does the piston fall?

A $0.040\ \mathrm{m}$

B 0.080 m

C 0.10 m

D 0.125 m

E 0.20 m

WORKED SOLUTION

The piston adds pressure

$$\frac{mg}{A} = \frac{50.0 \times 10}{0.0200} = 2.5 \times 10^4 \ \mathrm{Pa}.$$

So the final pressure is

$$1.00 \times 10^5 + 2.5 \times 10^4 = 1.25 \times 10^5 \ \mathrm{Pa}.$$

At constant temperature, pV is constant, so volume decreases by a factor

$$\frac{V_f}{V_i} = \frac{p_i}{p_f} = \frac{1.00}{1.25} = 0.80.$$

The cross-sectional area is constant, so the height also decreases by a factor of 0.80:

$$h_f = 0.80 \times 0.50 = 0.40 \ \mathrm{m}.$$

Therefore the piston falls by

$$0.50 - 0.40 = 0.10 \ \mathrm{m}.$$

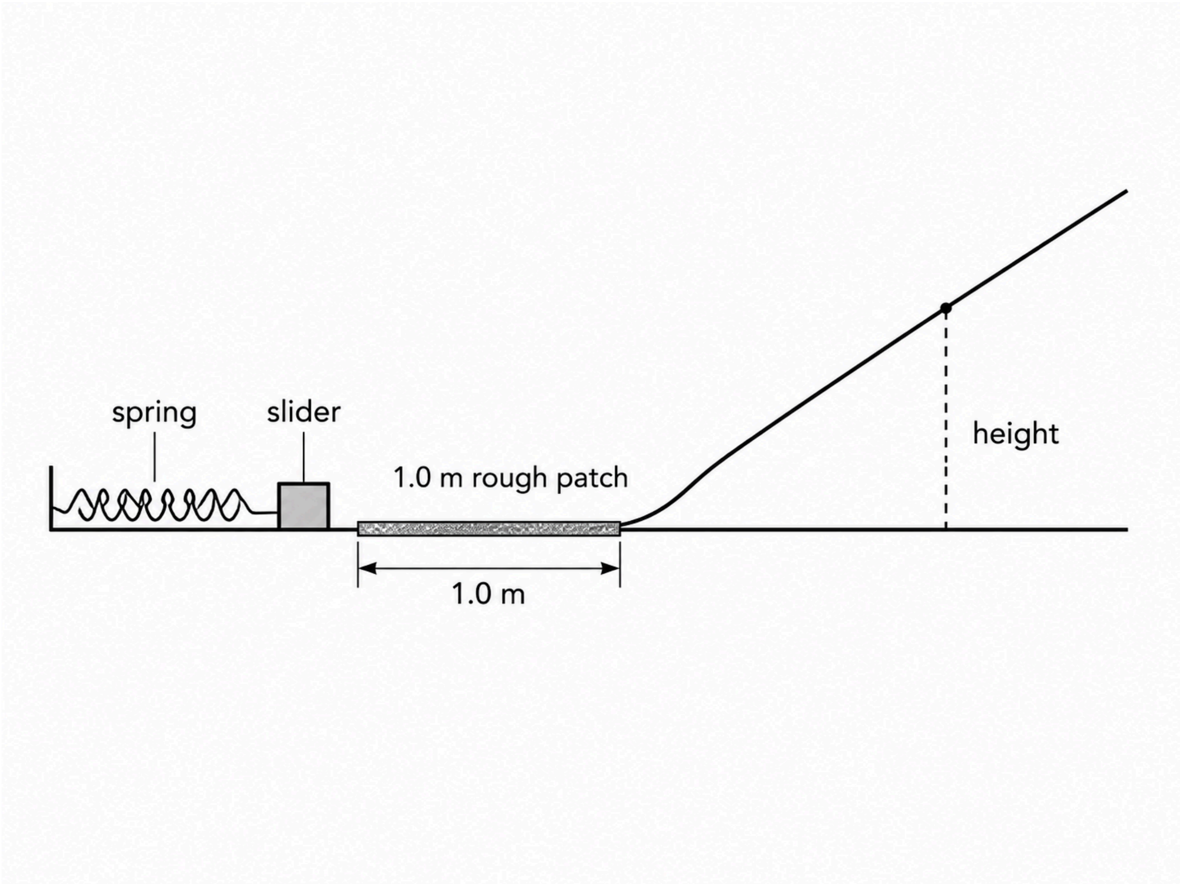
P-27

Springs, energy scaling and friction

Answer C

QUESTION 27

A small slider of mass $30\ \mathrm{g}$ is at rest near the bottom of a frictionless slope and in contact with a light uncompressed spring as shown.



In one test, compressing the spring by $5.0\ \mathrm{cm}$ sends the $30\ \mathrm{g}$ slider up the slope to a maximum vertical height of $20\ \mathrm{cm}$.

The slider is then replaced by a $20\ \mathrm{g}$ slider. The spring is compressed by $15\ \mathrm{cm}$. During the motion, the slider must first cross a rough horizontal patch of length $1.0\ \mathrm{m}$ on which a constant resistive force of $0.20\ \mathrm{N}$ acts. Elsewhere resistance is negligible.

To what maximum vertical height does the new slider rise? Take $g = 10\ \mathrm{N\ kg^{-1}}$.

A $120\ \mathrm{cm}$

B $150\ \mathrm{cm}$

C 170 cm

D 180 cm

E 270 cm

WORKED SOLUTION

From the first test, spring energy equals gain in gravitational potential energy:

$$\frac{1}{2}k(0.050)^2 = 0.030 \times 10 \times 0.20.$$

So

$$0.00125k = 0.060 \Rightarrow k = 48 \text{ N m}^{-1}.$$

With the new slider and compression 0.15 m, spring energy is

$$E_s = \frac{1}{2} \times 48 \times (0.15)^2 = 0.54 \text{ J}.$$

Work done against the rough patch is

$$W_f = 0.20 \times 1.0 = 0.20 \text{ J}.$$

So energy left for gravitational potential energy is

$$0.54 - 0.20 = 0.34 \text{ J}.$$

For the $20 \text{ g} = 0.020 \text{ kg}$ slider,

$$mgh = 0.34 \Rightarrow h = \frac{0.34}{0.020 \times 10} = 1.7 \text{ m} = 170 \text{ cm}.$$

LEVEL 2 · HARDER THAN ESAT

Chemistry

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Chemistry Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

C-01

Atomic structure and isotopes

Answer A

QUESTION 1

An ion X^- contains 18 electrons and 18 neutrons.

A sample of element X contains only two isotopes: the isotope described above and another isotope whose mass number is 2 greater.

The relative atomic mass of X in this sample is 35.6.

What percentage of naturally occurring X atoms are the heavier isotope?

A 30%

B 20%

C 25%

D 35%

E 40%

F 60%

WORKED SOLUTION

Because X^- has 18 electrons, neutral X has 17 electrons and therefore 17 protons.

The stated isotope therefore has mass number

$$17 + 18 = 35.$$

The two isotopes in the sample are 35 and 37. Let the fraction of the heavier isotope be p. Then

$$35(1 - p) + 37p = 35.6.$$

So

$$35 + 2p = 35.6 \Rightarrow p = 0.30.$$

The heavier isotope has abundance $\boxed{30\%}$, so the answer is **A**.

C-02 Precipitation and dissolved ions Answer **D**

QUESTION 2

$25.0\ \mathrm{cm}^3$ of $0.120\ \mathrm{mol\ dm}^{-3}$ aluminium sulfate solution is mixed with $10.0\ \mathrm{cm}^3$ of $0.300\ \mathrm{mol\ dm}^{-3}$ barium chloride solution.

Barium sulfate is insoluble. Assume all other substances remain completely dissociated in solution.

What is the **total amount, in moles, of dissolved ions remaining after precipitation is complete**?

- **A** 0.006
- **B** 0.009
- **C** 0.012
- **D** 0.018
- **E** 0.015

F 0.021

WORKED SOLUTION

Moles of $Al_2(SO_4)_3$:

$$0.120 \times 0.0250 = 0.00300.$$

This gives

$$0.00600 \text{ mol } Al^{3+}, \qquad 0.00900 \text{ mol } SO_4^{2-}.$$

Moles of $BaCl_2$:

$$0.300 \times 0.0100 = 0.00300.$$

This gives 0.00300 mol Ba^{2+} and 0.00600 mol Cl^-.

The precipitation is

$$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4(s).$$

All Ba^{2+} is removed, leaving

$$0.00900 - 0.00300 = 0.00600 \text{ mol } SO_4^{2-}.$$

Total dissolved ions remaining:

$$0.00600 + 0.00600 + 0.00600 = \boxed{0.0180 \text{ mol}}.$$

The answer is **D**.

C-03 Structure and bonding Answer **F**

QUESTION 3

The properties of five substances are shown.

substance	melting point	conducts when solid	conducts when molten	soluble in water
P	high	no	yes	yes
Q	very high	yes	yes	no
R	very high	no	no	no
S	moderate	yes	yes	no
T	very low	no	no	no

Which row gives the most likely structural types of **P, Q and R**, respectively?

A ionic, metallic, simple molecular

B giant covalent, metallic, ionic

C ionic, ionic, giant covalent

D metallic, graphite-like giant covalent, ionic

E simple molecular, metallic, giant covalent

F ionic, metallic, giant covalent

WORKED SOLUTION

P has the characteristic pattern of an ionic substance: high melting point, no conduction as a solid, conduction when molten and possible water solubility.

Q has a very high melting point, conducts in both the solid and molten states and is insoluble in water, consistent with a metallic structure.

R has a very high melting point and no mobile charge carriers, consistent with a giant covalent structure such as silicon dioxide.

Therefore the answer is **F**.

C-04

Visible chemical changes

Answer H

QUESTION 4

The following pairs of aqueous substances are mixed separately.

1. silver nitrate and potassium bromide
2. chlorine and potassium iodide
3. hydrochloric acid and sodium hydroxide, with no indicator present
4. sodium nitrate and potassium chloride

Which mixtures produce a **visible chemical change**?

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

E 1, 2 and 3 only

F 1, 2 and 4 only

G all four

H 1 and 2 only

WORKED SOLUTION

Mixture 1 forms a precipitate of silver bromide, so there is a visible change.

Mixture 2 undergoes halogen displacement:

$$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2,$$

giving a colour change.

Mixture 3 undergoes neutralisation, but without an indicator there is no necessary visible change. Mixture 4 leaves all ions in solution.

Therefore **1 and 2 only** are visible, answer **H**.

C-05

Reactivity and extraction

Answer **C**

QUESTION 5

Metal M has the following properties.

- It does not react rapidly with cold water.
- It reacts with steam.
- Its oxide can be reduced by carbon.
- It displaces copper from aqueous copper(II) sulfate.

Which could be M?

A aluminium

B calcium

C iron

D copper

E magnesium

F potassium

WORKED SOLUTION

The metal must be reactive enough to react with steam and displace copper, but it must lie below carbon because its oxide can be reduced by carbon.

Iron satisfies all of these conditions. Aluminium and magnesium are too reactive for their oxides to be reduced by carbon under the standard extraction model;

calcium and potassium are far more reactive; copper cannot displace itself.

The answer is **C: iron**.

C-06

Oxidising agents

Answer G

QUESTION 6

In which reactions is the **first reactant written** acting as an oxidising agent?

1. $$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$$
2. $$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$
3. $$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$
4. $$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$$

A 1 only

B 1 and 3 only

C 1 and 4 only

D 2 and 3 only

E 2, 3 and 4 only

F all four

G 1, 3 and 4 only

WORKED SOLUTION

An oxidising agent is itself reduced.

1. Chlorine changes from oxidation state 0 to −1, so Cl_2 is reduced and is an oxidising agent.

2. Zinc changes from 0 to +2, so it is oxidised and acts as a reducing agent.

3. Manganese changes from +4 in MnO_2 to +2 in $MnCl_2$, so MnO_2 is reduced.

4. $Fe^{3+} \rightarrow Fe^{2+}$ is reduction.

Thus the first reactant is an oxidising agent in **1, 3 and 4**, answer **G**.

C-07

Balancing an unfamiliar redox reaction

Answer **B**

QUESTION 7

A divalent metal M reacts with nitric acid to form:

- $M(NO_3)_2$,
- water,
- a single oxide of nitrogen X.

Exactly 0.075 mol of M reacts with 0.300 mol of nitric acid. No other products are formed.

What is the empirical formula of X?

A N_2O

B NO_2

C NO

D N_2O_3

E N_2O_5

WORKED SOLUTION

0.075 mol of M forms 0.075 mol of $M(NO_3)_2$, using

$$2(0.075) = 0.150 \text{ mol N atoms}$$

in nitrate.

Total nitrogen supplied is 0.300 mol, so nitrogen in X is

$$0.300 - 0.150 = 0.150 \text{ mol N atoms.}$$

The 0.300 mol of nitric acid supplies 0.300 mol of H atoms, so 0.150 mol of water is formed.

Total oxygen supplied is

$$3(0.300) = 0.900 \text{ mol O atoms.}$$

Oxygen used in nitrate is $6(0.075) = 0.450$ mol, and oxygen used in water is 0.150 mol. Therefore oxygen in X is

$$0.900 - 0.450 - 0.150 = 0.300.$$

Thus

$$N : O = 0.150 : 0.300 = 1 : 2,$$

so $X = \boxed{NO_2}$. The answer is **B**.

C-08

Acid-base mixture and pH

Answer G

QUESTION 8

50.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ hydrochloric acid is mixed with 150 cm^3 of $0.0125 \text{ mol dm}^{-3}$ calcium hydroxide solution.

Which range contains the pH of the final solution?

A $0 < \text{pH} < 1$

B $1 < \text{pH} < 2$

C $3 < \text{pH} < 7$

D $\text{pH} = 7$

E $7 < \mathrm{pH} < 12$

F $12 < \mathrm{pH} < 14$

G $2 < \mathrm{pH} < 3$

WORKED SOLUTION

Moles of H^+:

$$0.100(0.0500) = 0.00500.$$

Moles of $Ca(OH)_2$:

$$0.0125(0.150) = 0.001875.$$

This supplies

$$2(0.001875) = 0.00375 \text{ mol } OH^-.$$

Hydrogen ions are in excess:

$$0.00500 - 0.00375 = 0.00125 \text{ mol } H^+.$$

Total volume is 0.200 dm^3, so

$$[H^+] = \frac{0.00125}{0.200} = 0.00625 \text{ mol dm}^{-3}.$$

This concentration lies between 10^{-3} and 10^{-2} mol dm^{-3}. Therefore the pH lies between 2 and 3:

$$\boxed{2 < \mathrm{pH} < 3}.$$

The answer is **G**.

C-09

Deducing properties of an acid

Answer E

QUESTION 9

An acid solution has

$$\text{concentration} = 0.300\ \text{mol dm}^{-3}$$

and

$$\text{mass concentration} = 29.4\ \text{g dm}^{-3}.$$

A $25.0\ \text{cm}^3$ sample is completely neutralised by $45.0\ \text{cm}^3$ of $0.500\ \text{mol dm}^{-3}$ sodium hydroxide. The acid also reacts with magnesium to produce hydrogen.

Which statements can be deduced?

1. The relative molecular mass of the acid is 98.
2. One mole of the acid can supply three moles of H^+ in complete neutralisation.
3. The acid is a strong acid.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

The molar mass is

$$M_r = \frac{29.4}{0.300} = 98,$$

so statement 1 is true.

Moles of acid in $25.0\ cm^3$:

$$0.300(0.0250) = 0.00750.$$

Moles of $NaOH$:

$$0.500(0.0450) = 0.0225.$$

The ratio is

$$\frac{0.0225}{0.00750} = 3,$$

so statement 2 is true.

Reaction with magnesium does not show whether the acid is fully ionised in water, so statement 3 cannot be deduced.

The answer is **E: 1 and 2 only**.

C-10

Back titration and purity

Answer F

QUESTION 10

A $2.10\ g$ impure sample of magnesium carbonate contains impurities that do not react with acids.

The sample is reacted with $100\ cm^3$ of $0.500\ mol\ dm^{-3}$ hydrochloric acid.

The excess hydrochloric acid then requires $20.0\ cm^3$ of $0.500\ mol\ dm^{-3}$ sodium hydroxide for complete neutralisation.

$$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$$

$$M_r(MgCO_3) = 84$$

What is the percentage by mass of magnesium carbonate in the original sample?

A 40%

B 50%

C 60%

D 70%

E 90%

F 80%

WORKED SOLUTION

Initial HCl:

$$0.500(0.100) = 0.0500 \text{ mol.}$$

HCl remaining is equal to the amount of NaOH used:

$$0.500(0.0200) = 0.0100 \text{ mol.}$$

Therefore HCl consumed by $MgCO_3$ is

$$0.0500 - 0.0100 = 0.0400 \text{ mol.}$$

From the 1:2 ratio,

$$n(MgCO_3) = 0.0200 \text{ mol.}$$

Mass of pure $MgCO_3$:

$$0.0200(84) = 1.68 \text{ g.}$$

Hence

$$\% \text{ purity} = \frac{1.68}{2.10} \times 100 = \boxed{80\%}.$$

The answer is **F**.

C-11

Molecular formula from combustion

Answer C

QUESTION 11

A compound contains carbon, hydrogen and oxygen only.

Complete combustion of 0.880 g of the compound produces 1.760 g of CO_2 and 0.720 g of H_2O.

The relative molecular mass of the compound is 88.

$$A_r: \quad C = 12, \ H = 1, \ O = 16$$

What is the molecular formula?

- **A** CH_2O
- **B** C_2H_4O
- **C** $C_4H_8O_2$
- **D** $C_2H_4O_2$
- **E** $C_3H_6O_2$
- **F** C_4H_8O

WORKED SOLUTION

Carbon:

$$n(CO_2) = \frac{1.760}{44} = 0.0400,$$

so there are 0.0400 mol C atoms.

Hydrogen:

$$n(H_2O) = \frac{0.720}{18} = 0.0400,$$

so there are 0.0800 mol H atoms.

Mass of carbon is $0.0400(12) = 0.480$ g; mass of hydrogen is 0.0800 g. Therefore oxygen mass is

$$0.880 - 0.480 - 0.080 = 0.320 \text{ g},$$

giving

$$n(O) = \frac{0.320}{16} = 0.0200.$$

The ratio is

$$C : H : O = 0.040 : 0.080 : 0.020 = 2 : 4 : 1.$$

Empirical formula: C_2H_4O, mass 44. Since $88 = 2(44)$, the molecular formula is

$$\boxed{C_4H_8O_2}.$$

The answer is **C**.

C-12 Composition of air Answer **B**

QUESTION 12

A sample of dry air contains exactly 30.0 mol of gas.

Gas X contributes 3.78×10^{24} molecules to the sample.

Gas Y, if isolated at room temperature and pressure, would occupy $561.6\ \text{dm}^3$.

Take $N_A = 6.00 \times 10^{23}$ and molar gas volume at rtp as $24.0\ \text{dm}^3\ \text{mol}^{-1}$.

Which row is consistent with the approximate composition of dry air?

option	X	Y	remaining gases / mol
A	N_2	O_2	0.30
B	O_2	N_2	0.30
C	Ar	N_2	6.30
D	O_2	Ar	23.4
E	N_2	Ar	6.30

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

For X:

$$n = \frac{3.78 \times 10^{24}}{6.00 \times 10^{23}} = 6.30 \text{ mol.}$$

That is

$$\frac{6.30}{30.0} = 0.21 = 21\%,$$

so X is oxygen.

For Y:

$$n = \frac{561.6}{24.0} = 23.4 \text{ mol,}$$

which is

$$\frac{23.4}{30.0} = 0.78 = 78\%,$$

so Y is nitrogen.

Remaining gases:

$$30.0 - 6.30 - 23.4 = 0.30 \text{ mol.}$$

Therefore the correct row is **B**.

C-13

Two-dimensional chromatography

Answer **E**

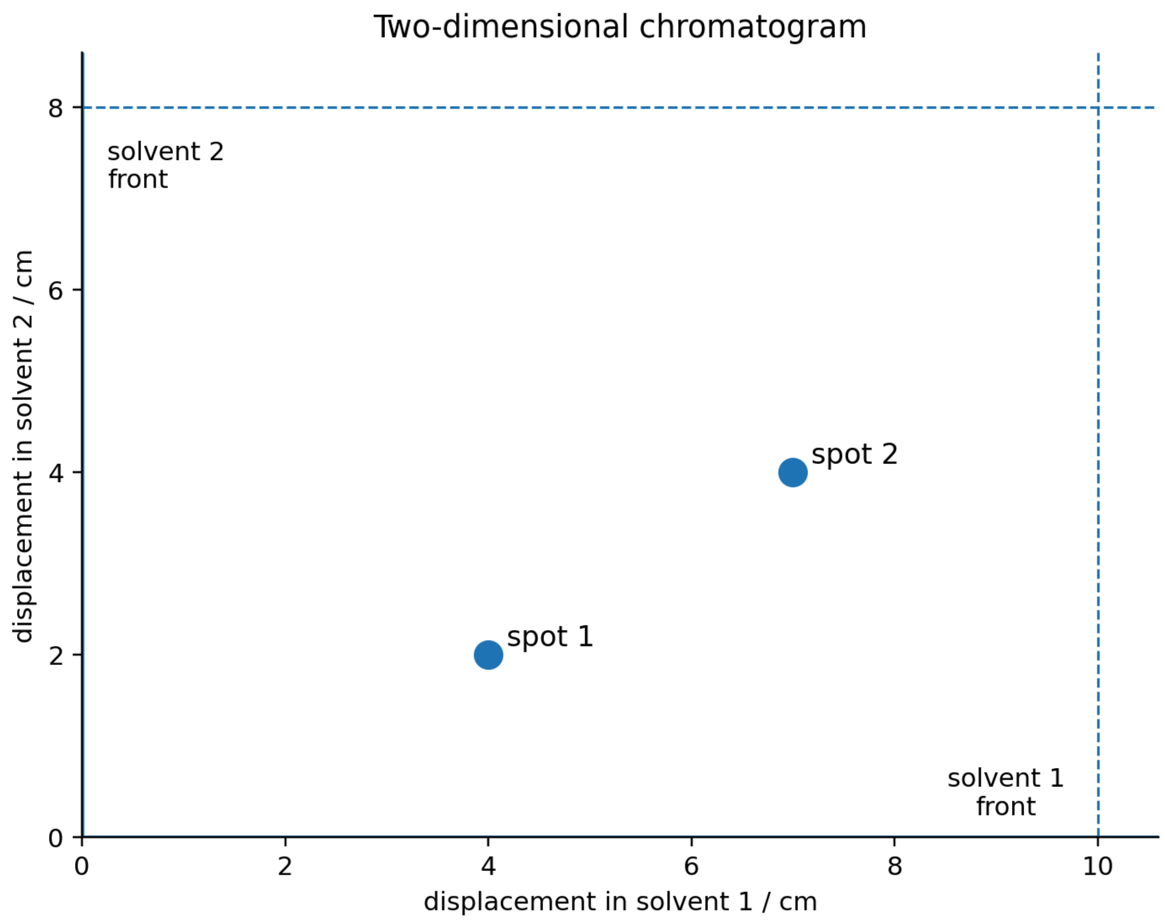
QUESTION 13

Four compounds have the following R_f values in two solvents.

compound	R_f in solvent 1	R_f in solvent 2
A	0.20	0.75
B	0.40	0.25
C	0.40	0.75
D	0.70	0.50

A mixture is analysed by two-dimensional chromatography. In the first direction, solvent 1 travels 10.0 cm. The paper is dried, rotated through 90°, and solvent 2 travels 8.0 cm.

The final chromatogram is shown.



Which compounds were present?

A A and B

B A and C

C B and C

D C and D

E B and D

WORKED SOLUTION

For spot 1:

$$R_{f,1} = \frac{4.0}{10.0} = 0.40, \qquad R_{f,2} = \frac{2.0}{8.0} = 0.25.$$

This identifies compound B.

For spot 2:

$$R_{f,1} = \frac{7.0}{10.0} = 0.70, \qquad R_{f,2} = \frac{4.0}{8.0} = 0.50.$$

This identifies compound D.

Therefore the mixture contained **B and D**, answer **E**.

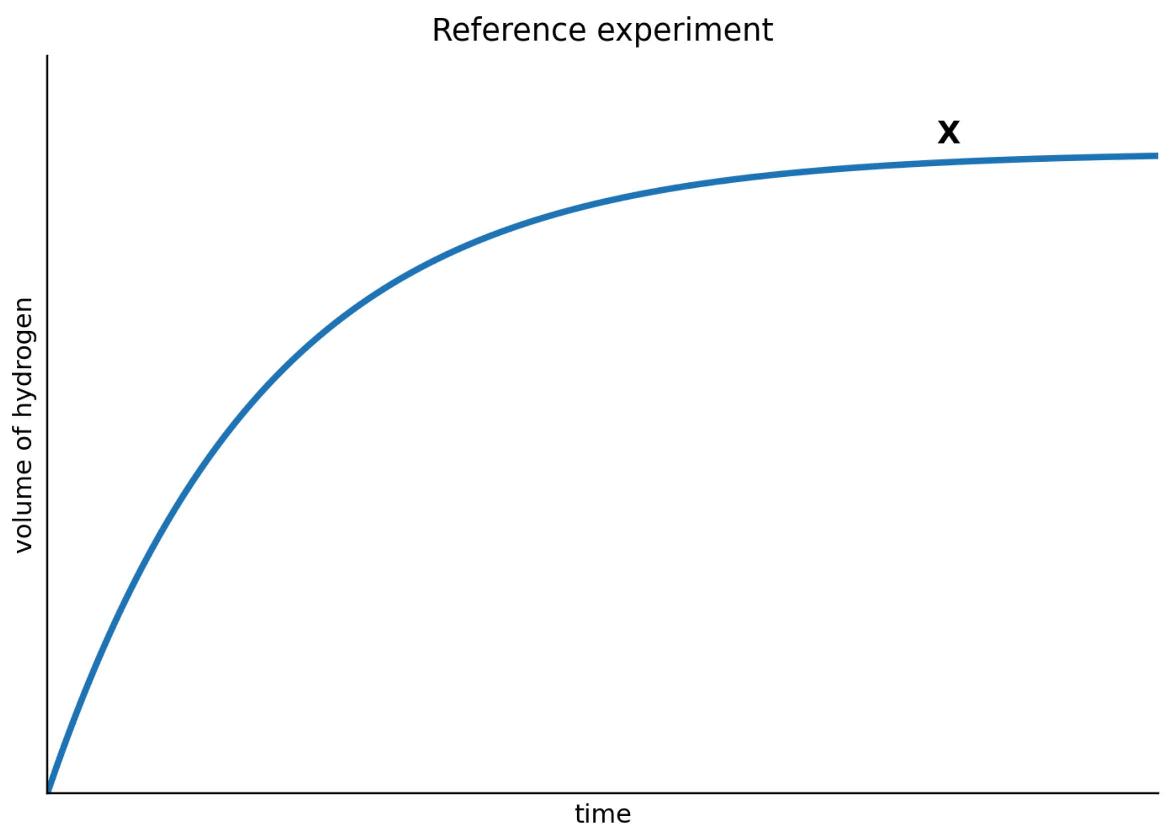
C-14

Rate and limiting reagent

Answer **A**

QUESTION 14

Curve X is obtained when $0.240\ g$ of magnesium granules react with $100\ cm^3$ of $0.100\ mol\ dm^{-3}$ hydrochloric acid.



The experiment is repeated using:

- the same mass of magnesium, but powdered;
- $100\ cm^3$ of $0.150\ mol\ dm^{-3}$ hydrochloric acid.

$$A_r(Mg) = 24$$

Which statement correctly compares the second curve with X?

option	initial gradient	final volume of H_2
A	greater	$1.5 \times X$
B	greater	same as X
C	same	$1.5 \times X$
D	smaller	$1.5 \times X$
E	greater	$2.0 \times X$

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Moles of magnesium:

$$\frac{0.240}{24} = 0.0100.$$

Complete reaction would need 0.0200 mol HCl.

First experiment:

$$n(HCl) = 0.100(0.100) = 0.0100 \text{ mol},$$

so acid is limiting.

Second experiment:

$$n(HCl) = 0.150(0.100) = 0.0150 \text{ mol},$$

which is still limiting. Therefore the final hydrogen amount is larger by

$$\frac{0.0150}{0.0100} = 1.5.$$

The second experiment also has a higher acid concentration and a larger magnesium surface area, so its initial rate is greater.

The correct row is **A**.

C-15

Equilibrium from observations

Answer C

QUESTION 15

Gaseous reactants P and Q establish equilibrium with gaseous products R and S.

During establishment of equilibrium from the reactants:

- for every 0.10 mol of P consumed,
- 0.20 mol of Q is consumed,
- 0.10 mol of R is formed.

Increasing pressure changes the rate at which equilibrium is reached but does **not** change the equilibrium yield.

Increasing temperature reduces the equilibrium yield of products.

Which equation is consistent with all the observations?

A $P + 2Q \rightleftharpoons R + S \qquad \Delta H < 0$

B $P + 2Q \rightleftharpoons 2R + S \qquad \Delta H > 0$

C $P + 2Q \rightleftharpoons R + 2S \qquad \Delta H < 0$

D $2P + Q \rightleftharpoons R + 2S \qquad \Delta H < 0$

E $P + 2Q \rightleftharpoons R + 2S \qquad \Delta H > 0$

F $P + Q \rightleftharpoons R + S \qquad \Delta H < 0$

WORKED SOLUTION

The observed mole changes give the ratio

$$P : Q : R = 1 : 2 : 1.$$

So the equation must begin

$$P + 2Q \rightleftharpoons R + \cdots$$

There are three gaseous reactant particles. Because pressure does not shift the equilibrium yield, the product side must also contain three gaseous particles. Thus the stoichiometry is

$$P + 2Q \rightleftharpoons R + 2S.$$

Increasing temperature reduces product yield, so the forward reaction is exothermic: $\Delta H < 0$.

The answer is **C**.

C-16

Bond energies

Answer **D**

QUESTION 16

Ethene reacts with hydrogen:

$$C_2H_4 + H_2 \rightarrow C_2H_6$$

The enthalpy change is $-125\ \text{kJ mol}^{-1}$.

Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H–H	436
C=C	612
C–C	348

What is the mean C–H bond energy implied by these data?

A 356 kJ mol^{-1}

B 388 kJ mol^{-1}

C 436 kJ mol^{-1}

D 413 kJ mol^{-1}

E 475 kJ mol^{-1}

WORKED SOLUTION

Relative to the reactants, hydrogenation breaks one C=C bond and one H–H bond, and forms one C–C bond and two additional C–H bonds.

Let the mean C–H bond energy be x. Then

$$-125 = (612 + 436) - (348 + 2x).$$

So

$$-125 = 700 - 2x \Rightarrow 2x = 825 \Rightarrow x = 412.5.$$

Therefore the closest value is $\boxed{413 \text{ kJ mol}^{-1}}$, answer **D**.

C-17

Calorimetry and limiting reagent

Answer C

QUESTION 17

$40.0\ \mathrm{cm}^3$ of $1.50\ \mathrm{mol\ dm}^{-3}$ hydrochloric acid is mixed with $60.0\ \mathrm{cm}^3$ of $0.500\ \mathrm{mol\ dm}^{-3}$ sodium hydroxide.

The temperature rises by $3.4°\mathrm{C}$.

Assume:

- the solution volumes are additive;
- density of the final solution = $1.0\ \mathrm{g\ cm}^{-3}$;
- specific heat capacity = $4.2\ \mathrm{J\ g}^{-1}\ °\mathrm{C}^{-1}$;
- only 85% of the energy released by the reaction is retained by the solution.

What is the molar enthalpy change for the neutralisation reaction?

A $-28\ \mathrm{kJ\ mol}^{-1}$

B $-40\ \mathrm{kJ\ mol}^{-1}$

C $-56\ \mathrm{kJ\ mol}^{-1}$

D $-48\ \mathrm{kJ\ mol}^{-1}$

E $-66\ \mathrm{kJ\ mol}^{-1}$

WORKED SOLUTION

Moles HCl:

$$1.50(0.0400) = 0.0600.$$

Moles NaOH:

$$0.500(0.0600) = 0.0300.$$

NaOH is limiting, so 0.0300 mol is neutralised.

The solution mass is 100 g. Energy retained:

$$q = mc\Delta T = 100(4.2)(3.4) = 1428\ \mathrm{J}.$$

This is 85% of the energy released, so

$$q_{\mathrm{released}} = \frac{1428}{0.85} = 1680\ \mathrm{J} = 1.680\ \mathrm{kJ}.$$

Per mole:

$$\frac{1.680}{0.0300} = 56.0\ \mathrm{kJ\ mol^{-1}}.$$

Neutralisation is exothermic, so $\boxed{-56.0\ \mathrm{kJ\ mol^{-1}}}$, answer **C**.

C-18

Metal extraction

Answer **B**

QUESTION 18

An iron ore contains 80.0% by mass of hydrated iron(III) oxide,

$$Fe_2O_3 \cdot 3H_2O.$$

The formula $Fe_2O_3 \cdot 3H_2O$ means that each formula unit contains one Fe_2O_3 unit together with three H_2O units.

Assume all the iron in the hydrated oxide can be extracted.

$$A_r: \quad Fe = 56,\ O = 16,\ H = 1$$

What is the minimum mass of ore required to produce 56.0 tonnes of iron?

A 100 tonnes

B 134 tonnes

C 112 tonnes

D 125 tonnes

E 160 tonnes

F 214 tonnes

WORKED SOLUTION

The relative formula mass of $Fe_2O_3 \cdot 3H_2O$ is

$$2(56) + 3(16) + 3(18) = 214.$$

Each 214 mass units contain 112 mass units of iron, so the iron fraction in pure hydrated oxide is $112/214$.

The ore is only 80.0% hydrated oxide, so the iron fraction of the ore is

$$0.80\left(\frac{112}{214}\right).$$

If m tonnes of ore are needed,

$$m \cdot 0.80\left(\frac{112}{214}\right) = 56.0.$$

Thus

$$m = 133.75 \text{ tonnes} \approx \boxed{134 \text{ tonnes}}.$$

The answer is **B**.

C-19

Electrolysis and concentration

Answer E

QUESTION 19

$250\ cm^3$ of $1.00\ mol\ dm^{-3}$ copper(II) sulfate solution is electrolysed using inert electrodes.

After some time, $6.40\ g$ of copper has been deposited at the cathode.

Water is then added so that the solution volume is again exactly $250\ \mathrm{cm}^3$.

$$A_r(\mathrm{Cu}) = 64$$

What is the final concentration of Cu^{2+}?

A $0.20\ \mathrm{mol\ dm}^{-3}$

B $0.40\ \mathrm{mol\ dm}^{-3}$

C $0.50\ \mathrm{mol\ dm}^{-3}$

D $0.75\ \mathrm{mol\ dm}^{-3}$

E $0.60\ \mathrm{mol\ dm}^{-3}$

F $0.90\ \mathrm{mol\ dm}^{-3}$

WORKED SOLUTION

Initial amount of Cu^{2+}:

$$1.00(0.250) = 0.250\ \mathrm{mol}.$$

Copper deposited:

$$n(\mathrm{Cu}) = \frac{6.40}{64} = 0.100\ \mathrm{mol}.$$

Therefore 0.100 mol of Cu^{2+} has been removed, leaving

$$0.250 - 0.100 = 0.150\ \mathrm{mol}.$$

At a final volume of $0.250\ \mathrm{dm}^3$,

$$[\mathrm{Cu}^{2+}] = \frac{0.150}{0.250} = \boxed{0.600\ \mathrm{mol\ dm}^{-3}}.$$

The answer is **E**.

C-20

Organic structure from constraints

Answer **B**

QUESTION 20

A hydrocarbon X:

- produces $120\ \mathrm{dm}^3$ of carbon dioxide when one mole is completely burned;
- reacts with exactly one mole of hydrogen per mole of X;
- undergoes addition polymerisation;
- has a terminal C=C bond;
- forms a **branched saturated hydrocarbon** on hydrogenation.

At rtp, one mole of gas occupies $24\ \mathrm{dm}^3$.

The candidate structures are shown as condensed structural formulae.

structure	condensed structural formula
A	$CH_2 = CHCH_2CH_2CH_3$
B	$CH_2 = C(CH_3)CH_2CH_3$
C	$CH_3CH = CHCH_2CH_3$
D	$CH_3C(CH_3) = CHCH_3$
E	cyclopentane, $(CH_2)_5$ ring

Which could be X?

A structure A

B structure B

C structure C

D structure D

E structure E

WORKED SOLUTION

The carbon dioxide volume corresponds to

$$\frac{120}{24} = 5\ \text{mol}\ CO_2,$$

so X contains five carbon atoms.

Reaction with exactly one mole of hydrogen indicates one C=C bond. Addition polymerisation also requires an alkene.

The double bond must be terminal, leaving A or B. Hydrogenating A gives unbranched pentane, whereas hydrogenating B gives branched 2-methylbutane.

Therefore X is **structure B**.

C-21 Multiple double bonds | Answer A

QUESTION 21

A hydrocarbon contains two carbon-carbon double bonds.

A $0.100\ \text{cm}^3$ sample has density $0.820\ \text{g cm}^{-3}$ and relative molecular mass 82.

What is the minimum volume of $0.250\ \text{mol dm}^{-3}$ bromine solution required to react completely with the hydrocarbon?

A $8.0\ \text{cm}^3$

B $2.0\ \text{cm}^3$

C $4.0\ \text{cm}^3$

D $6.0\ \text{cm}^3$

E $16.0\ cm^3$

WORKED SOLUTION

Mass of hydrocarbon:

$$0.100(0.820) = 0.0820\text{ g.}$$

Moles:

$$\frac{0.0820}{82} = 0.00100\text{ mol.}$$

There are two C=C bonds, so each mole consumes two moles of bromine. Thus

$$n(Br_2) = 0.00200\text{ mol.}$$

Required volume:

$$V = \frac{0.00200}{0.250} = 0.00800\ dm^3 = \boxed{8.00\ cm^3}.$$

The answer is **A**.

C-22 Carboxylic acid and esterification Answer **D**

QUESTION 22

A saturated monocarboxylic acid X reacts with propan-1-ol to form an ester Y.

$$M_r(Y) = 130$$

and

$$M_r(\text{propan-1-ol}) = 60.$$

What is the molecular formula of X?

A $C_2H_4O_2$

B $C_3H_6O_2$

C $C_5H_{10}O_2$

D $C_4H_8O_2$

E $C_6H_{12}O_2$

WORKED SOLUTION

Esterification gives

$$\text{acid} + \text{alcohol} \rightarrow \text{ester} + H_2O.$$

Therefore

$$M_r(X) + 60 = 130 + 18,$$

so

$$M_r(X) = 88.$$

A saturated monocarboxylic acid has formula $C_nH_{2n}O_2$, so

$$12n + 2n + 32 = 88.$$

Thus

$$14n = 56 \Rightarrow n = 4.$$

The formula is $\boxed{C_4H_8O_2}$, answer **D**.

C-23 Combustion and cracking Answer **F**

QUESTION 23

$20\ \text{cm}^3$ of a gaseous alkane X requires $190\ \text{cm}^3$ of oxygen for complete combustion.

Both gas volumes are measured at the same temperature and pressure.

X can crack according to

$$X \rightarrow C_2H_4 + Y$$

with one molecule of Y formed for each molecule of X.

What is Y?

A C_2H_6

B C_3H_6

C C_3H_8

D C_4H_8

E C_5H_{10}

F C_4H_{10}

WORKED SOLUTION

For an alkane C_nH_{2n+2}, complete combustion requires

$$n + \frac{2n + 2}{4} = \frac{3n + 1}{2}$$

moles of oxygen per mole of alkane.

The gas-volume ratio is

$$\frac{190}{20} = 9.5.$$

Hence

$$\frac{3n + 1}{2} = 9.5 \Rightarrow 3n + 1 = 19 \Rightarrow n = 6.$$

So $X = C_6H_{14}$. Balancing

$$C_6H_{14} \rightarrow C_2H_4 + Y$$

gives

$$Y = \boxed{C_4H_{10}}.$$

The answer is **F**.

C-24

Water treatment and gas quantities

Answer **D**

QUESTION 24

A water-treatment plant processes $480\,000\ \mathrm{dm}^3$ of water each day.

Chlorine is added at a rate of $0.355\ \mathrm{mg\ dm}^{-3}$.

Assume all the chlorine is supplied as $Cl_2(g)$.

$$A_r(Cl) = 35.5$$

and one mole of gas occupies $24.0\ \mathrm{dm}^3$ at rtp.

What volume of chlorine gas is used each day?

- **A** $24.0\ \mathrm{dm}^3$
- **B** $36.0\ \mathrm{dm}^3$
- **C** $48.0\ \mathrm{dm}^3$
- **D** $57.6\ \mathrm{dm}^3$
- **E** $71.0\ \mathrm{dm}^3$
- **F** $96.0\ \mathrm{dm}^3$

WORKED SOLUTION

Mass of chlorine used:

$$480000(0.355) = 170400 \text{ mg} = 170.4 \text{ g}.$$

Since $M_r(Cl_2) = 71$,

$$n(Cl_2) = \frac{170.4}{71} = 2.40 \text{ mol}.$$

At rtp,

$$V = 2.40(24.0) = \boxed{57.6 \text{ dm}^3}.$$

The answer is **D**.

C-25

Qualitative analysis

Answer F

QUESTION 25

An aqueous salt X gives:

- a **green precipitate** on addition of aqueous sodium hydroxide;
- a **white precipitate** on addition of acidified aqueous barium chloride.

Which is the most likely identity of X?

A $FeCl_2$

B $Fe_2(SO_4)_3$

C $CuSO_4$

D $MgSO_4$

E Na_2SO_4

F $FeSO_4$

WORKED SOLUTION

A green hydroxide precipitate identifies Fe^{2+}.

A white precipitate with acidified barium chloride identifies sulfate ions, SO_4^{2-}.

The salt must therefore be $\boxed{FeSO_4}$, answer **F**.

C-26

Multi-stage gas stoichiometry

Answer **E**

QUESTION 26

Potassium chlorate decomposes completely:

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

All the oxygen produced is then used to oxidise excess carbon monoxide:

$$2CO + O_2 \rightarrow 2CO_2.$$

A $24.5\ \mathrm{g}$ sample of pure potassium chlorate is used.

$$M_r(KClO_3) = 122.5$$

and one mole of gas occupies $24.0\ \mathrm{dm}^3$ at rtp.

What volume of carbon dioxide is ultimately formed?

- **A** $4.8\ \mathrm{dm}^3$
- **B** $7.2\ \mathrm{dm}^3$
- **C** $9.6\ \mathrm{dm}^3$
- **D** $12.0\ \mathrm{dm}^3$
- **E** $14.4\ \mathrm{dm}^3$

F $18.0\ dm^3$

WORKED SOLUTION

Moles of potassium chlorate:

$$\frac{24.5}{122.5} = 0.200\ \text{mol.}$$

From

$$2KClO_3 \rightarrow 3O_2,$$

the oxygen amount is

$$0.200\left(\frac{3}{2}\right) = 0.300\ \text{mol.}$$

Then

$$O_2 : CO_2 = 1 : 2,$$

so

$$n(CO_2) = 0.600\ \text{mol.}$$

Volume:

$$0.600(24.0) = \boxed{14.4\ dm^3}.$$

The answer is **E**.

C-27

Mixture analysis and back titration

Answer A

QUESTION 27

A $3.00\ g$ mixture contains only magnesium and magnesium oxide.

It is reacted with $100\ cm^3$ of $2.00\ mol\ dm^{-3}$ hydrochloric acid.

The excess hydrochloric acid requires $20.0\ cm^3$ of $1.00\ mol\ dm^{-3}$ sodium hydroxide for complete neutralisation.

$$Mg + 2HCl \rightarrow MgCl_2 + H_2$$

$$MgO + 2HCl \rightarrow MgCl_2 + H_2O$$

$$A_r(Mg) = 24, \qquad M_r(MgO) = 40.$$

What volume of hydrogen gas is produced at rtp?

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

A 0.90 dm^3

B 0.45 dm^3

C 0.60 dm^3

D 0.75 dm^3

E 1.20 dm^3

F 1.80 dm^3

WORKED SOLUTION

Initial HCl:

$$0.100(2.00) = 0.200 \text{ mol}.$$

Excess HCl:

$$0.0200(1.00) = 0.0200 \text{ mol}.$$

So HCl consumed by the mixture is

$$0.200 - 0.0200 = 0.180 \text{ mol}.$$

Both Mg and MgO consume two moles HCl per mole, so

$$n(Mg) + n(MgO) = \frac{0.180}{2} = 0.0900.$$

Let $n(Mg) = x$ and $n(MgO) = y$. Then

$$x + y = 0.0900$$

and

$$24x + 40y = 3.00.$$

Using $x = 0.0900 - y$:

$$24(0.0900 - y) + 40y = 3.00$$

$$2.16 + 16y = 3.00 \Rightarrow y = 0.0525.$$

Hence

$$x = 0.0900 - 0.0525 = 0.0375.$$

Only magnesium metal produces hydrogen, in a 1:1 ratio, so

$$n(H_2) = 0.0375.$$

Therefore

$$V(H_2) = 0.0375(24.0) = \boxed{0.900\ \text{dm}^3}.$$

The answer is **A**.

Compilation record: Level 2 uses Engineering Mock Test 5 for Mathematics/Physics and the corresponding Harder than ESAT Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.